EduQG pilot - izvorni materijal

Izvorni pasusi (hl_context) iz kojih su izvedena ekspertska pitanja evaluiranog pilot skupa. 149 jedinstvenih pasusa iz 12 OpenStax udzbenika. Sluzi kao ulaz za generisanje sopstvenih pitanja i poredjenje sa EduQG pitanjima.

american government

Poglavlje 4

In either case , the amendment indicates that government officials are required to apply for and receive a search warrant prior to a search or seizure ; this warrant is a legal document , signed by a judge , allowing police to search and / or seize persons or property . Since the 1960s , however , the Supreme Court has issued a series of rulings limiting the warrant requirement in situations where a person can be said to lack a " reasonable expectation of privacy " outside the home . Police can also search and / or seize people or property without a warrant if the owner or renter consents to the search , if there is a reasonable expectation that evidence may be destroyed or tampered with before a warrant can be issued (i . e . , exigent circumstances) , or if the items in question are in plain view of government officials . The text of the Fourth Amendment is as follows :

Poglavlje 7

In 2015 , Oregon made news when it took the concept of Motor Voter further . When citizens turn eighteen , the state now automatically registers most of them using driver ' s license and state identification information . When a citizen moves , the voter rolls are updated when the license is updated . While this policy has been controversial , with some arguing that private information may become public or that Oregon is moving toward mandatory voting , automatic registration is consistent with the state ' s efforts to increase registration and turnout . 16

Poglavlje 8

Television combined the best attributes of radio and pictures and changed media forever . The first official broadcast in the United States was President Franklin Roosevelt ' s speech at the opening of the 1939 World ' s Fair in New York . The public did not immediately begin buying televisions , but coverage of World War II changed their minds . CBS reported on war events and included pictures and maps that enhanced the news for viewers . By the 1950s , the price of television sets had dropped , more televisions stations were being created , and advertisers were buying up spots .

Poglavlje 10

Interest groups also include association s , which are typically groups of institutions that join with others , often within the same trade or industry (trade associations) , and have similar concerns . The American Beverage Association 10 includes Coca-Cola , Red Bull North America , ROCKSTAR , and Kraft Foods . Despite the fact that these companies are competitors , they have common interests related to the manufacturing , bottling , and distribution of beverages , as well as the regulation of their business activities . The logic is that there is strength in numbers , and if members can lobby for tax breaks or eased regulations for an entire industry , they may all benefit . These common goals do not , however , prevent individual association members from employing in-house lobbyists or contract lobbying firms to represent their own business or organization as well . Indeed , many members of associations are competitors who also seek representation individually before the legislature .

Poglavlje 11

These aren ' t geographical features or large infrastructure projects . Rather , they are racially gerrymandered congressional districts . Their strange shapes are the product of careful district restructuring organized around the goal of enhancing the votes of minority groups . The alligator-mouth District 4 in Illinois , for example , was drawn to bring a number of geographically autonomous Latino groups in Illinois together in the same congressional district .

Poglavlje 13

A justice ' s decisions are influenced by how he or she defines his role as a jurist , with some justices believing strongly in judicial activism , or the need to defend individual rights and liberties , and they aim to stop actions and laws by other branches of government that they see as infringing on these rights . A judge or justice who views the role with an activist lens is more likely to use his or her judicial power to broaden personal liberty , justice , and equality . Still others believe in judicial restraint , which leads them to defer decisions (and thus policymaking) to the elected branches of government and stay focused on a narrower interpretation of the Bill of Rights . These justices are less likely to strike down actions or laws as unconstitutional and are less likely to focus on the expansion of individual liberties . While it is typically the case that liberal actions are described as unnecessarily activist , conservative decisions can be activist as well .

Poglavlje 16

While Social Security was designed to provide cash payments to sustain the aged and disabled , Medicare and Medicaid were intended to ensure that vulnerable populations have access to health care . Medicare , like Social Security , is an entitlement program funded through payroll taxes . Its purpose is to make sure that senior citizens and retirees have access to low-cost health care they might not otherwise have , because most U . S . citizens get their health insurance through their employers . Medicare provides three major forms of coverage : a guaranteed insurance benefit that helps cover major hospitalization , fee-based supplemental coverage that retirees can use to lower costs for doctor visits and other health expenses , and a prescription drug benefit . Medicare faces many of the same long-term challenges as Social Security , due to the same demographic shifts . Medicare also faces the problem that health care costs are rising significantly faster than inflation . In 2014 , Medicare cost the federal government almost $ 597 billion . 16

Poglavlje 17

While key White House , executive , and legislative leaders monitor and regularly weigh in on foreign policy matters , the fact is that individual representatives and senators do so much less often . Unless there is a foreign policy crisis , legislators in Congress tend to focus on domestic matters , mainly because there is not much to be gained with their constituents by pursuing foreign policy matters . 11 Domestic policy matters resonate more strongly with the voters at home . A sluggish economy , increasing health care costs , and crime matter more to them than U . S . policy toward North Korea , for example . In an open-ended Gallup poll question from early 2016 about the " most important problem " in the United States , fewer than 15 percent of respondents named a foreign policy topic (half of those respondents mentioned immigration) . These results suggest that foreign policy is not at the top of many voters ' minds . In the end , legislators must be responsive to constituents in order to be good representatives and to achieve reelection . 12

anatomy and physiology

Poglavlje 2

Although electrons do not follow rigid orbits a set distance away from the atom ' s nucleus , they do tend to stay within certain regions of space called electron shells . An electron shell is a layer of electrons that encircle the nucleus at a distinct energy level . The atoms of the elements found in the human body have from one to five electron shells , and all electron shells hold eight electrons except the first shell , which can only hold two . This configuration of electron shells is the same for all atoms . The precise number of shells depends on the number of electrons in the atom . Hydrogen and helium have just one and two electrons , respectively . If you take a look at the periodic table of the elements , you will notice that hydrogen and helium are placed alone on either sides of the top row ; they are the only elements that have just one electron shell (Figure 2.7) . A second shell is necessary to hold the electrons in all elements larger than hydrogen and helium .

Poglavlje 3

Much like the processes of DNA replication and transcription , translation consists of three main stages : initiation , elongation , and termination . Initiation takes place with the binding of a ribosome to an mRNA transcript . The elongation stage involves the recognition of a tRNA anticodon with the next mRNA codon in the sequence . Once the anticodon and codon sequences are bound (remember , they are complementary base pairs) , the tRNA presents its amino acid cargo and the growing polypeptide strand is attached to this next amino acid . This attachment takes place with the assistance of various enzymes and requires energy . The tRNA molecule then releases the mRNA strand , the mRNA strand shifts one codon over in the ribosome , and the next appropriate tRNA arrives with its matching anticodon . This process continues until the final codon on the mRNA is reached which provides a " stop " message that signals termination of translation and triggers the release of the complete , newly synthesized protein . Thus , a gene within the DNA molecule is transcribed into mRNA , which is then translated into a protein product (Figure 3.29) . Remember that many of a cell ' s ribosomes are found associated with the rough ER , and carry out the synthesis of proteins destined for the Golgi apparatus . Ribosomal RNA (rRNA) is a type of RNA that , together with proteins , composes the structure of the ribosome . Ribosomes exist in the cytoplasm as two distinct components , a small and a large subunit . When an mRNA molecule is ready to be translated , the two subunits come together and attach to the mRNA . The ribosome provides a substrate for translation , bringing together and aligning the mRNA molecule with the molecular " translators " that must decipher its code . The other major requirement for protein synthesis is the translator molecules that physically " read " the mRNA codons . Transfer RNA (tRNA) is a type of RNA that ferries the appropriate corresponding amino acids to the ribosome , and attaches each new amino acid to the last , building the polypeptide chain one-by-one . Thus tRNA transfers specific amino acids from the cytoplasm to a growing polypeptide . The tRNA molecules must be able to recognize the codons on mRNA and match them with the correct amino acid . The tRNA is modified for this function . On one end of its structure is a binding site for a specific amino acid . On the other end is a base sequence that matches the codon specifying its particular amino acid . This sequence of three bases on the tRNA molecule is called an anticodon . For example , a tRNA responsible for shuttling the amino acid glycine contains a binding site for glycine on one end . On the other end it contains an anticodon that complements the glycine codon (GGA is a codon for glycine , and so the tRNAs anticodon would read CCU) . Equipped with its particular cargo and matching anticodon , a tRNA molecule can read its recognized mRNA codon and bring the corresponding amino acid to the growing chain (Figure 3.28) . Before the mRNA molecule leaves the nucleus and proceeds to protein synthesis , it is modified in a number of ways . For this reason , it is often called a pre-mRNA at this stage . For example , your DNA , and thus complementary mRNA , contains long regions called non-coding regions that do not code for amino acids . Their function is still a mystery , but the process called splicing removes these non-coding regions from the pre-mRNA transcript (Figure 3.27) . A spliceosome - a structure composed of various proteins and other molecules - attaches to the mRNA and " splices " or cuts out the non-coding regions . The removed segment of the transcript is called an intron . The remaining exons are pasted together . An exon is a segment of RNA that remains after splicing . Interestingly , some introns that are removed from mRNA are not always non-coding . When different coding regions of mRNA are spliced out , different variations of the protein will eventually result , with differences in structure and function . This process results in a much larger variety of possible proteins and protein functions . When the mRNA transcript is ready , it travels out of the nucleus and into the cytoplasm . DNA is housed within the nucleus , and protein synthesis takes place in the cytoplasm , thus there must be some sort of intermediate messenger that leaves the nucleus and manages protein synthesis . This intermediate messenger is messenger RNA (mRNA) , a single-stranded nucleic acid that carries a copy of the genetic code for a single gene out of the nucleus and into the cytoplasm where it is used to produce proteins .

Poglavlje 4

Apocrine secretion accumulates near the apical portion of the cell . That portion of the cell and its secretory contents pinch off from the cell and are released . Apocrine sweat glands in the axillary and genital areas release fatty secretions that local bacteria break down ; this causes body odor . Both merocrine and apocrine glands continue to produce and secrete their contents with little damage caused to the cell because the nucleus and golgi regions remain intact after secretion . In contrast , the process of holocrine secretion involves the rupture and destruction of the entire gland cell . The cell accumulates its secretory products and releases them only when it bursts . New gland cells differentiate from cells in the surrounding tissue to replace those lost by secretion . The sebaceous glands that produce the oils on the skin and hair are holocrine glands / cells (Figure 4.11) .

The progressive impact of aging on the body varies considerably among individuals , but Studies indicate , however , that exercise and healthy lifestyle choices can slow down the deterioration of the body that comes with old age .

Adipose tissue consists mostly of fat storage cells , with little extracellular matrix (Figure 4.13) . A large number of capillaries allow rapid storage and mobilization of lipid molecules . White adipose tissue is most abundant . It can appear yellow and owes its color to carotene and related pigments from plant food . White fat contributes mostly to lipid storage and can serve as insulation from cold temperatures and mechanical injuries . White adipose tissue can be found protecting the kidneys and cushioning the back of the eye . Brown adipose tissue is more common in infants , hence the term " baby fat . " In adults , there is a reduced amount of brown fat and it is found mainly in the neck and clavicular regions of the body . The many mitochondria in the cytoplasm of brown adipose tissue help explain its efficiency at metabolizing stored fat . Brown adipose tissue is thermogenic , meaning that as it breaks down fats , it releases metabolic heat , rather than producing adenosine triphosphate (ATP) , a key molecule used in metabolism . Areolar tissue shows little specialization . It contains all the cell types and fibers previously described and is distributed in a random , web-like fashion . It fills the spaces between muscle fibers , surrounds blood and lymph vessels , and supports organs in the abdominal cavity . Areolar tissue underlies most epithelia and represents the connective tissue component of epithelial membranes , which are described further in a later section . Connective tissues perform many functions in the body , but most importantly , they support and connect other tissues ; from the connective tissue sheath that surrounds muscle cells , to the tendons that attach muscles to bones , and to the skeleton that supports the positions of the body . Protection is another major function of connective tissue , in the form of fibrous capsules and bones that protect delicate organs and , of course , the skeletal system . Specialized cells in connective tissue defend the body from microorganisms that enter the body . Transport of fluid , nutrients , waste , and chemical messengers is ensured by specialized fluid connective tissues , such as blood and lymph . Adipose cells store surplus energy in the form of fat and contribute to the thermal insulation of the body .

Poglavlje 6

While bones are increasing in length , they are also increasing in diameter ; growth in diameter can continue even after longitudinal growth ceases . This is called appositional growth . Osteoclasts resorb old bone that lines the medullary cavity , while osteoblasts , via intramembranous ossification , produce new bone tissue beneath the periosteum . The erosion of old bone along the medullary cavity and the deposition of new bone beneath the periosteum not only increase the diameter of the diaphysis but also increase the diameter of the medullary cavity . This process is called modeling .

Vitamin K also supports bone mineralization and may have a synergistic role with vitamin D in the regulation of bone growth . Green leafy vegetables are a good source of vitamin K .

The epiphyseal plate is the area of growth in a long bone . It is a layer of hyaline cartilage where ossification occurs in immature bones . On the epiphyseal side of the epiphyseal plate , cartilage is formed . On the diaphyseal side , cartilage is ossified , and the diaphysis grows in length . The epiphyseal plate is composed of four zones of cells and activity (Figure 6.18) . The reserve zone is the region closest to the epiphyseal end of the plate and contains small chondrocytes within the matrix . These chondrocytes do not participate in bone growth but secure the epiphyseal plate to the osseous tissue of the epiphysis .

Poglavlje 7

Ribs 1 - 7 are classified as true ribs (vertebrosternal ribs) . The costal cartilage from each of these ribs attaches directly to the sternum . Ribs 8 - 12 are called false ribs (vertebrochondral ribs) . The costal cartilages from these ribs do not attach directly to the sternum . For ribs 8 - 10 , the costal cartilages are attached to the cartilage of the next higher rib . Thus , the cartilage of rib 10 attaches to the cartilage of rib 9 , rib 9 then attaches to rib 8 , and rib 8 is attached to rib 7 . The last two false ribs (11 - 12) are also called floating ribs (vertebral ribs) . These are short ribs that do not attach to the sternum at all . Instead , their small costal cartilages terminate within the musculature of the lateral abdominal wall . 7.5 Embryonic Development of the Axial Skeleton Learning Objectives By the end of this section , you will be able to :

Poglavlje 9

The articular capsule of the elbow is thin on its anterior and posterior aspects , but is thickened along its outside margins by strong intrinsic ligaments . These ligaments prevent side-to-side movements and hyperextension . On the medial side is the triangular ulnar collateral ligament . This arises from the medial epicondyle of the humerus and attaches to the medial side of the proximal ulna . The strongest part of this ligament is the anterior portion , which resists hyperextension of the elbow . The ulnar collateral ligament may be injured by frequent , forceful extensions of the forearm , as is seen in baseball pitchers . Reconstructive surgical repair of this ligament is referred to as Tommy John surgery , named for the former major league pitcher who was the first person to have this treatment . The lateral side of the elbow is supported by the radial collateral ligament . This arises from the lateral epicondyle of the humerus and then blends into the lateral side of the annular ligament . The annular ligament encircles the head of the radius . This ligament supports the head of the radius as it articulates with the radial notch of the ulna at the proximal radioulnar joint . This is a pivot joint that allows for rotation of the radius during supination and pronation of the forearm .

Abduction and adduction motions occur within the coronal plane and involve medial-lateral motions of the limbs , fingers , toes , or thumb . Abduction moves the limb laterally away from the midline of the body , while adduction is the opposing movement that brings the limb toward the body or across the midline . For example , abduction is raising the arm at the shoulder joint , moving it laterally away from the body , while adduction brings the arm down to the side of the body . Similarly , abduction and adduction at the wrist moves the hand away from or toward the midline of the body . Spreading the fingers or toes apart is also abduction , while bringing the fingers or toes together is adduction . For the thumb , abduction is the anterior movement that brings the thumb to a 90 ° perpendicular position , pointing straight out from the palm . Adduction moves the thumb back to the anatomical position , next to the index finger . Abduction and adduction movements are seen at condyloid , saddle , and ball-and-socket joints (see Figure 9.12 e) .

Poglavlje 10

All living cells have membrane potentials , or electrical gradients across their membranes . The inside of the membrane is usually around - 60 to - 90 mV , relative to the outside . This is referred to as a cell ' s membrane potential . Neurons and muscle cells can use their membrane potentials to generate electrical signals . They do this by controlling the movement of charged particles , called ions , across their membranes to create electrical currents . This is achieved by opening and closing specialized proteins in the membrane called ion channels . Although the currents generated by ions moving through these channel proteins are very small , they form the basis of both neural signaling and muscle contraction . Both neurons and skeletal muscle cells are electrically excitable , meaning that they are able to generate action potentials . An action potential is a special type of electrical signal that can travel along a cell membrane as a wave . This allows a signal to be transmitted quickly and faithfully over long distances . Although the term excitation-contraction coupling confuses or scares some students , it comes down to this : for a skeletal muscle fiber to contract , its membrane must first be " excited " - in other words , it must be stimulated to fire an action potential . The muscle fiber action potential , which sweeps along the sarcolemma as a wave , is " coupled " to the actual contraction through the release of calcium ions (Ca + +) from the SR . Once released , the Ca + + interacts with the shielding proteins , forcing them to move aside so that the actin-binding sites are available for attachment by myosin heads . The myosin then pulls the actin filaments toward the center , shortening the muscle fiber . In skeletal muscle , this sequence begins with signals from the somatic motor division of the nervous system . In other words , the " excitation " step in skeletal muscles is always triggered by signaling from the nervous system (Figure 10.6) . The motor neurons that tell the skeletal muscle fibers to contract originate in the spinal cord , with a smaller number located in the brainstem for activation of skeletal muscles of the face , head , and neck . These neurons have long processes , called axons , which are specialized to transmit action potentials long distances - in this case , all the way from the spinal cord to the muscle itself (which may be up to three feet away) . The axons of multiple neurons bundle together to form nerves , like wires bundled together in a cable . Signaling begins when a neuronal action potential travels along the axon of a motor neuron , and then along the individual branches to terminate at the NMJ . At the NMJ , the axon terminal releases a chemical messenger , or neurotransmitter , called acetylcholine (ACh) . The ACh molecules diffuse across a minute space called the synaptic cleft and bind to ACh receptors located within the motor end-plate of the sarcolemma on the other side of the synapse . Once ACh binds , a channel in the ACh receptor opens and positively charged ions can pass through into the muscle fiber , causing it to depolarize , meaning that the membrane potential of the muscle fiber becomes less negative (closer to zero .

) As the membrane depolarizes , another set of ion channels called voltage-gated sodium channels are triggered to open . Sodium ions enter the muscle fiber , and an action potential rapidly spreads (or " fires ") along the entire membrane to initiate excitation-contraction coupling . Things happen very quickly in the world of excitable membranes (just think about how quickly you can snap your fingers as soon as you decide to do it) . Immediately following depolarization of the membrane , it repolarizes , re-establishing the negative membrane potential . Meanwhile , the ACh in the synaptic cleft is degraded by the enzyme acetylcholinesterase (AChE) so that the ACh cannot rebind to a receptor and reopen its channel , which would cause unwanted extended muscle excitation and contraction . Propagation of an action potential along the sarcolemma is the excitation portion of excitation-contraction coupling . Recall that this excitation actually triggers the release of calcium ions (Ca + +) from its storage in the cell ' s SR . For the action potential to reach the membrane of the SR , there are periodic invaginations in the sarcolemma , called T-tubules (" T " stands for " transverse ") . You will recall that the diameter of a muscle fiber can be up to 100 ? m , so these T-tubules ensure that the membrane can get close to the SR in the sarcoplasm . The arrangement of a T-tubule with the membranes of SR on either side is called a triad (Figure 10.7) . The triad surrounds the cylindrical structure called a myofibril , which contains actin and myosin . The T-tubules carry the action potential into the interior of the cell , which triggers the opening of calcium channels in the membrane of the adjacent SR , causing Ca + + to diffuse out of the SR and into the sarcoplasm . It is the arrival of Ca + + in the sarcoplasm that initiates contraction of the muscle fiber by its contractile units , or sarcomeres . 10.3 Muscle Fiber Contraction and Relaxation Learning Objectives By the end of this section , you will be able to :

Poglavlje 11

In anatomical terminology , chewing is called mastication . Muscles involved in chewing must be able to exert enough pressure to bite through and then chew food before it is swallowed (Figure 11.10 and Table 11.4) . The masseter muscle is the main muscle used for chewing because it elevates the mandible (lower jaw) to close the mouth , and it is assisted by the temporalis muscle , which retracts the mandible . You can feel the temporalis move by putting your fingers to your temple as you chew . Muscles of the Lower Jaw

Poglavlje 12

A region of the cortex is specialized for sending signals down to the spinal cord for movement . The upper motor neuron is in this region , called the precentral gyrus of the frontal cortex , which has an axon that extends all the way down the spinal cord . At the level of the spinal cord at which this axon makes a synapse , a graded potential occurs in the cell membrane of a lower motor neuron . This second motor neuron is responsible for causing muscle fibers to contract . In the manner described in the chapter on muscle tissue , an action potential travels along the motor neuron axon into the periphery . The axon terminates on muscle fibers at the neuromuscular junction . Acetylcholine is released at this specialized synapse , which causes the muscle action potential to begin , following a large potential known as an end plate potential . When the lower motor neuron excites the muscle fiber , it contracts . All of this occurs in a fraction of a second , but this story is the basis of how the nervous system functions . Career Connection Neurophysiologist

Poglavlje 15

In a similar fashion , another aspect of the cardiovascular system is primarily under sympathetic control . Blood pressure is partially determined by the contraction of smooth muscle in the walls of blood vessels . These tissues have adrenergic receptors that respond to the release of norepinephrine from postganglionic sympathetic fibers by constricting and increasing blood pressure . The hormones released from the adrenal medulla - epinephrine and norepinephrine - will also bind to these receptors . Those hormones travel through the bloodstream where they can easily interact with the receptors in the vessel walls . The parasympathetic system has no significant input to the systemic blood vessels , so the sympathetic system determines their tone . Another example is in the control of pupillary size (Figure 15.9) . The afferent branch responds to light hitting the retina . Photoreceptors are activated , and the signal is transferred to the retinal ganglion cells that send an action potential along the optic nerve into the diencephalon . If light levels are low , the sympathetic system sends a signal out through the upper thoracic spinal cord to the superior cervical ganglion of the sympathetic chain . The postganglionic fiber then projects to the iris , where it releases norepinephrine onto the radial fibers of the iris (a smooth muscle) . When those fibers contract , the pupil dilates - increasing the amount of light hitting the retina . If light levels are too high , the parasympathetic system sends a signal out from the Edinger - Westphal nucleus through the oculomotor nerve . This fiber synapses in the ciliary ganglion in the posterior orbit . The postganglionic fiber then projects to the iris , where it releases ACh onto the circular fibers of the iris - another smooth muscle . When those fibers contract , the pupil constricts to limit the amount of light hitting the retina .

Poglavlje 16

The general senses are distributed throughout the body , relying on nervous tissue incorporated into various organs . Somatic senses are incorporated mostly into the skin , muscles , or tendons , whereas the visceral senses come from nervous tissue incorporated into the majority of organs such as the heart or stomach . The somatic senses are those that usually make up the conscious perception of the how the body interacts with the environment . The visceral senses are most often below the limit of conscious perception because they are involved in homeostatic regulation through the autonomic nervous system . The facial and glossopharyngeal nerves are also responsible for the initiation of salivation . Neurons in the salivary nuclei of the medulla project through these two nerves as preganglionic fibers , and synapse in ganglia located in the head . The parasympathetic fibers of the facial nerve synapse in the pterygopalatine ganglion , which projects to the submandibular gland and sublingual gland . The parasympathetic fibers of the glossopharyngeal nerve synapse in the otic ganglion , which projects to the parotid gland . Salivation in response to food in the oral cavity is based on a visceral reflex arc within the facial or glossopharyngeal nerves . Other stimuli that stimulate salivation are coordinated through the hypothalamus , such as the smell and sight of food . The olfactory , optic , and vestibulocochlear nerves (cranial nerves I , II , and VIII) are dedicated to four of the special senses : smell , vision , equilibrium , and hearing , respectively . Taste sensation is relayed to the brain stem through fibers of the facial and glossopharyngeal nerves . The trigeminal nerve is a mixed nerve that carries the general somatic senses from the head , similar to those coming through spinal nerves from the rest of the body .

Poglavlje 18

Interleukins are another class of cytokine signaling molecules important in hemopoiesis . They were initially thought to be secreted uniquely by leukocytes and to communicate only with other leukocytes , and were named accordingly , but are now known to be produced by a variety of cells including bone marrow and endothelium . Researchers now suspect that interleukins may play other roles in body functioning , including differentiation and maturation of cells , producing immunity and inflammation . To date , more than a dozen interleukins have been identified , with others likely to follow . They are generally numbered IL - 1 , IL - 2 , IL - 3 , etc .

Albumin is the most abundant of the plasma proteins . Manufactured by the liver , albumin molecules serve as binding proteins - transport vehicles for fatty acids and steroid hormones . Recall that lipids are hydrophobic ; however , their binding to albumin enables their transport in the watery plasma . Albumin is also the most significant contributor to the osmotic pressure of blood ; that is , its presence holds water inside the blood vessels and draws water from the tissues , across blood vessel walls , and into the bloodstream . This in turn helps to maintain both blood volume and blood pressure . Albumin normally accounts for approximately 54 percent of the total plasma protein content , in clinical levels of 3.5 - 5.0 g / dL blood .

Poglavlje 19

Ventricular relaxation , or diastole , follows repolarization of the ventricles and is represented by the T wave of the ECG . It too is divided into two distinct phases and lasts approximately 430 ms .

Poglavlje 21

A macrophage is an irregularly shaped phagocyte that is amoeboid in nature and is the most versatile of the phagocytes in the body . Macrophages move through tissues and squeeze through capillary walls using pseudopodia . They not only participate in innate immune responses but have also evolved to cooperate with lymphocytes as part of the adaptive immune response . Macrophages exist in many tissues of the body , either freely roaming through connective tissues or fixed to reticular fibers within specific tissues such as lymph nodes . When pathogens breach the body ' s barrier defenses , macrophages are the first line of defense (Table 21.3) . They are called different names , depending on the tissue : Kupffer cells in the liver , histiocytes in connective tissue , and alveolar macrophages in the lungs .

Poglavlje 24

The digestion of proteins begins in the stomach . When protein-rich foods enter the stomach , they are greeted by a mixture of the enzyme pepsin and hydrochloric acid (HCl ; 0.5 percent) . The latter produces an environmental pH of 1.5 - 3.5 that denatures proteins within food . Pepsin cuts proteins into smaller polypeptides and their constituent amino acids . When the food-gastric juice mixture (chyme) enters the small intestine , the pancreas releases sodium bicarbonate to neutralize the HCl . This helps to protect the lining of the intestine . The small intestine also releases digestive hormones , including secretin and CCK , which stimulate digestive processes to break down the proteins further . Secretin also stimulates the pancreas to release sodium bicarbonate . The pancreas releases most of the digestive enzymes , including the proteases trypsin , chymotrypsin , and elastase , which aid protein digestion . Together , all of these enzymes break complex proteins into smaller individual amino acids (Figure 24.17) , which are then transported across the intestinal mucosa to be used to create new proteins , or to be converted into fats or acetyl CoA and used in the Krebs cycle .

Of the four major macromolecular groups (carbohydrates , lipids , proteins , and nucleic acids) that are processed by digestion , carbohydrates are considered the most common source of energy to fuel the body . They take the form of either complex carbohydrates , polysaccharides like starch and glycogen , or simple sugars (monosaccharides) like glucose and fructose . Sugar catabolism breaks polysaccharides down into their individual monosaccharides . Among the monosaccharides , glucose is the most common fuel for ATP production in cells , and as such , there are a number of endocrine control mechanisms to regulate glucose concentration in the bloodstream . Excess glucose is either stored as an energy reserve in the liver and skeletal muscles as the complex polymer glycogen , or it is converted into fat (triglyceride) in adipose cells (adipocytes) . Metabolic processes are constantly taking place in the body . Metabolism is the sum of all of the chemical reactions that are involved in catabolism and anabolism . The reactions governing the breakdown of food to obtain energy are called catabolic reactions . Conversely , anabolic reactions use the energy produced by catabolic reactions to synthesize larger molecules from smaller ones , such as when the body forms proteins by stringing together amino acids . Both sets of reactions are critical to maintaining life .

Poglavlje 25

Introduction The urinary system has roles you may be well aware of : cleansing the blood and ridding the body of wastes probably come to mind . However , there are additional , equally important functions played by the system . Take for example , regulation of pH , a function shared with the lungs and the buffers in the blood . Additionally , the regulation of blood pressure is a role shared with the heart and blood vessels . What about regulating the concentration of solutes in the blood ? Did you know that the kidney is important in determining the concentration of red blood cells ? Eighty-five percent of the erythropoietin (EPO) produced to stimulate red blood cell production is produced in the kidneys . The kidneys also perform the final synthesis step of vitamin D production , converting calcidiol to calcitriol , the active form of vitamin D . If the kidneys fail , these functions are compromised or lost altogether , with devastating effects on homeostasis . The affected individual might experience weakness , lethargy , shortness of breath , anemia , widespread edema (swelling) , metabolic acidosis , rising potassium levels , heart arrhythmias , and more . Each of these functions is vital to your well-being and survival . The urinary system , controlled by the nervous system , also stores urine until a convenient time for disposal and then provides the anatomical structures to transport this waste liquid to the outside of the body . Failure of nervous control or the anatomical structures leading to a loss of control of urination results in a condition called incontinence . This chapter will help you to understand the anatomy of the urinary system and how it enables the physiologic functions critical to homeostasis . It is best to think of the kidney as a regulator of plasma makeup rather than simply a urine producer . As you read each section , ask yourself this question : " What happens if this does not work ? " This question will help you to understand how the urinary system maintains homeostasis and affects all the other systems of the body and the quality of one ' s life . Interactive Link

EPO is a 193 - amino acid protein that stimulates the formation of red blood cells in the bone marrow . The kidney produces 85 percent of circulating EPO ; the liver , the remainder . If you move to a higher altitude , the partial pressure of oxygen is lower , meaning there is less pressure to push oxygen across the alveolar membrane and into the red blood cell . One way the body compensates is to manufacture more red blood cells by increasing EPO production . If you start an aerobic exercise program , your tissues will need more oxygen to cope , and the kidney will respond with more EPO . If erythrocytes are lost due to severe or prolonged bleeding , or under produced due to disease or severe malnutrition , the kidneys come to the rescue by producing more EPO . Renal failure (loss of EPO production) is associated with anemia , which makes it difficult for the body to cope with increased oxygen demands or to supply oxygen adequately even under normal conditions . Anemia diminishes performance and can be life threatening . Introduction The urinary system has roles you may be well aware of : cleansing the blood and ridding the body of wastes probably come to mind . However , there are additional , equally important functions played by the system . Take for example , regulation of pH , a function shared with the lungs and the buffers in the blood . Additionally , the regulation of blood pressure is a role shared with the heart and blood vessels . What about regulating the concentration of solutes in the blood ? Did you know that the kidney is important in determining the concentration of red blood cells ? Eighty-five percent of the erythropoietin (EPO) produced to stimulate red blood cell production is produced in the kidneys . The kidneys also perform the final synthesis step of vitamin D production , converting calcidiol to calcitriol , the active form of vitamin D . If the kidneys fail , these functions are compromised or lost altogether , with devastating effects on homeostasis . The affected individual might experience weakness , lethargy , shortness of breath , anemia , widespread edema (swelling) , metabolic acidosis , rising potassium levels , heart arrhythmias , and more . Each of these functions is vital to your well-being and survival . The urinary system , controlled by the nervous system , also stores urine until a convenient time for disposal and then provides the anatomical structures to transport this waste liquid

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The loop of Henle consists of two sections : thick and thin descending and thin and thick ascending sections . The loops of cortical nephrons do not extend into the renal medulla very far , if at all . Juxtamedullary nephrons have loops that extend variable distances , some very deep into the medulla . The descending and ascending portions of the loop are highly specialized to enable recovery of much of the Na + and water that were filtered by the glomerulus . As the forming urine moves through the loop , the osmolarity will change from isosmotic with blood (about 278 - 300 mOsmol / kg) to both a very hypertonic solution of about 1200 mOsmol / kg and a very hypotonic solution of about 100 mOsmol / kg . These changes are accomplished by osmosis in the descending limb and active transport in the ascending limb . Solutes and water recovered from these loops are returned to the circulation by way of the vasa recta . In a dissected kidney , it is easy to identify the cortex ; it appears lighter in color compared to the rest of the kidney . All of the renal corpuscles as well as both the proximal convoluted tubules (PCTs) and distal convoluted tubules are found here . Some nephrons have a short loop of Henle that does not dip beyond the cortex . These nephrons are called cortical nephrons . About 15 percent of nephrons have long loops of Henle that extend deep into the medulla and are called juxtamedullary nephrons . 25.4 Microscopic Anatomy of the Kidney Learning Objectives By the end of this section , you will be able to :

Decreased blood pressure is also sensed by the granular cells in the afferent arteriole of the JGA . In response , the enzyme renin is released . You saw earlier in the chapter that renin activity leads to an almost immediate rise in blood pressure as activated angiotensin II produces vasoconstriction . The rise in pressure is sustained by the aldosterone effects initiated by angiotensin II ; this includes an increase in Na + retention and water volume . As an aside , late in the menstrual cycle , progesterone has a modest influence on water retention . Due to its structural similarity to aldosterone , progesterone binds to the aldosterone receptor in the collecting duct of the kidney , causing the same , albeit weaker , effect on Na + and water retention .

Poglavlje 27

As a woman reaches the age of menopause , depletion of the number of viable follicles in the ovaries due to atresia affects the hormonal regulation of the menstrual cycle . During the years leading up to menopause , there is a decrease in the levels of the hormone inhibin , which normally participates in a negative feedback loop to the pituitary to control the production of FSH . The menopausal decrease in inhibin leads to an increase in FSH . The presence of FSH stimulates more follicles to grow and secrete estrogen . Because small , secondary follicles also respond to increases in FSH levels , larger numbers of follicles are stimulated to grow ; however , most undergo atresia and die . Eventually , this process leads to the depletion of all follicles in the ovaries , and the production of estrogen falls off dramatically . It is primarily the lack of estrogens that leads to the symptoms of menopause .

Poglavlje 28

Colostrum is secreted during the first 48 - 72 hours postpartum . Only a small volume of colostrum is produced - approximately 3 ounces in a 24 - hour period - but it is sufficient for the newborn in the first few days of life . Colostrum is rich with immunoglobulins , which confer gastrointestinal , and also likely systemic , immunity as the newborn adjusts to a nonsterile environment .

biology

Poglavlje 4

A prokaryote is a simple , mostly single-celled (unicellular) organism that lacks a nucleus , or any other membrane-bound organelle . We will shortly come to see that this is significantly different in eukaryotes . Prokaryotic DNA is found in a central part of the cell : the nucleoid (Figure 4.5) . Most prokaryotes have a peptidoglycan cell wall and many have a polysaccharide capsule (Figure 4.5) . The cell wall acts as an extra layer of protection , helps the cell maintain its shape , and prevents dehydration . The capsule enables the cell to attach to surfaces in its environment . Some prokaryotes have flagella , pili , or fimbriae . Flagella are used for locomotion . Pili are used to exchange genetic material during a type of reproduction called conjugation . Fimbriae are used by bacteria to attach to a host cell . Career Connection Microbiologist

Poglavlje 5

Factors That Affect Diffusion Molecules move constantly in a random manner , at a rate that depends on their mass , their environment , and the amount of thermal energy they possess , which in turn is a function of temperature . This movement accounts for the diffusion of molecules through whatever medium in which they are localized . A substance will tend to move into any space available to it until it is evenly distributed throughout it . After a substance has diffused completely through a space , removing its concentration gradient , molecules will still move around in the space , but there will be no net movement of the number of molecules from one area to another . This lack of a concentration gradient in which there is no net movement of a substance is known as dynamic equilibrium . While diffusion will go forward in the presence of a concentration gradient of a substance , several factors affect the rate of diffusion . Each separate substance in a medium , such as the extracellular fluid , has its own concentration gradient , independent of the concentration gradients of other materials . In addition , each substance will diffuse according to that gradient . Within a system , there will be different rates of diffusion of the different substances in the medium .

Poglavlje 7

Step 2 . In the second step of glycolysis , an isomerase converts glucose - 6 - phosphate into one of its isomers , fructose - 6 - phosphate . An isomerase is an enzyme that catalyzes the conversion of a molecule into one of its isomers . (This change from phosphoglucose to phosphofructose allows the eventual split of the sugar into two three-carbon molecules .) . Step 1 . The first step in glycolysis (Figure 7.6) is catalyzed by hexokinase , an enzyme with broad specificity that catalyzes the phosphorylation of six-carbon sugars . Hexokinase phosphorylates glucose using ATP as the source of the phosphate , producing glucose - 6 - phosphate , a more reactive form of glucose . This reaction prevents the phosphorylated glucose molecule from continuing to interact with the GLUT proteins , and it can no longer leave the cell because the negatively charged phosphate will not allow it to cross the hydrophobic interior of the plasma membrane .

Poglavlje 10

During telophase , the " distance phase , " the chromosomes reach the opposite poles and begin to decondense (unravel) , relaxing into a chromatin configuration . The mitotic spindles are depolymerized into tubulin monomers that will be used to assemble cytoskeletal components for each daughter cell . Nuclear envelopes form around the chromosomes , and nucleosomes appear within the nuclear area .

Animal cells Linear chromosomes exist in the nucleus . A mitotic spindle forms from the centrosomes . The nuclear envelope dissolves . Chromosomes attach to the mitotic spindle , which separates the chromosomes and elongates the cell . Microfilaments form a cleavage furrow that pinches the cell in two .

Poglavlje 16

Silencing genes through epigenetic mechanisms is also very common in cancer cells . There are characteristic modifications to histone proteins and DNA that are associated with silenced genes . In cancer cells , the DNA in the promoter region of silenced genes is methylated on cytosine DNA residues in CpG islands . Histone proteins that surround that region lack the acetylation modification that is present when the genes are expressed in normal cells . This combination of DNA methylation and histone deacetylation (epigenetic modifications that lead to gene silencing) is commonly found in cancer . When these modifications occur , the gene present in that chromosomal region is silenced . Increasingly , scientists understand how epigenetic changes are altered in cancer . Because these changes are temporary and can be reversed - for example , by preventing the action of the histone deacetylase protein that removes acetyl groups , or by DNA methyl transferase enzymes that add methyl groups to cytosines in DNA - it is possible to design new drugs and new therapies to take advantage of the reversible nature of these processes . Indeed , many researchers are testing how a silenced gene can be switched back on in a cancer cell to help re-establish normal growth patterns . This type of gene regulation is called epigenetic regulation . Epigenetic means " around genetics . " The changes that occur to the histone proteins and DNA do not alter the nucleotide sequence and are not permanent . Instead , these changes are temporary (although they often persist through multiple rounds of cell division) and alter the chromosomal structure (open or closed) as needed . A gene can be turned on or off depending upon the location and modifications to the histone proteins and DNA . If a gene is to be transcribed , the histone proteins and DNA are modified surrounding the chromosomal region encoding that gene . This opens the chromosomal region to allow access for RNA polymerase and other proteins , called transcription factors , to bind to the promoter region , located just upstream of the gene , and initiate transcription . If a gene is to remain turned off , or silenced , the histone proteins and DNA have different modifications that signal a closed chromosomal configuration . In this closed configuration , the RNA polymerase and transcription factors do not have access to the DNA and transcription cannot occur (Figure 16.7) .

Proto-oncogenes are positive cell-cycle regulators . When mutated , proto-oncogenes can become oncogenes and cause cancer . Overexpression of the oncogene can lead to uncontrolled cell growth . This is because oncogenes can alter transcriptional activity , stability , or protein translation of another gene that directly or indirectly controls cell growth . An example of an oncogene involved in cancer is a protein called myc . Myc is a transcription factor that is aberrantly activated in Burkett ' s Lymphoma , a cancer of the lymph system . Overexpression of myc transforms normal B cells into cancerous cells that continue to grow uncontrollably . High B-cell numbers can result in tumors that can interfere with normal bodily function . Patients with Burkett ' s lymphoma can develop tumors on their jaw or in their mouth that interfere with the ability to eat .

Poglavlje 17

The basic sequencing technique used in all modern day sequencing projects is the chain termination method (also known as the dideoxy method) , which was developed by Fred Sanger in the 1970s . The chain termination method involves DNA replication of a single-stranded template with the use of a primer and a regular deoxynucleotide (dNTP) , which is a monomer , or a single unit , of DNA . The primer and dNTP are mixed with a small proportion of fluorescently labeled dideoxynucleotides (ddNTPs) . The ddNTPs are monomers that are missing a hydroxyl group (- OH) at the site at which another nucleotide usually attaches to form a chain (Figure 17.13) . Each ddNTP is labeled with a different color of fluorophore . Every time a ddNTP is incorporated in the growing complementary strand , it terminates the process of DNA replication , which results in multiple short strands of replicated DNA that are each terminated at a different point during replication . When the reaction mixture is processed by gel electrophoresis after being separated into single strands , the multiple newly replicated DNA strands form a ladder because of the differing sizes . Because the ddNTPs are fluorescently labeled , each band on the gel reflects the size of the DNA strand and the ddNTP that terminated the reaction . The different colors of the fluorophore-labeled ddNTPs help identify the ddNTP incorporated at that position . Reading the gel on the basis of the color of each band on the ladder produces the sequence of the template strand (Figure 17.14) .

Poglavlje 20

If a characteristic is found in the ancestor of a group , it is considered a shared ancestral character because all of the organisms in the taxon or clade have that trait . The vertebrate in Figure 20.10 is a shared ancestral character . Now consider the amniotic egg characteristic in the same figure . Only some of the organisms in Figure 20.10 have this trait , and to those that do , it is called a shared derived character because this trait derived at some point but does not include all of the ancestors in the tree . The tricky aspect to shared ancestral and shared derived characters is the fact that these terms are relative . The same trait can be considered one or the other depending on the particular diagram being used . Returning to Figure 20.10 , note that the amniotic egg is a shared ancestral character for the Amniota clade , while having hair is a shared derived character for some organisms in this group . These terms help scientists distinguish between clades in the building of phylogenetic trees . Choosing the Right Relationships

Poglavlje 21

A virus attaches to a specific receptor site on the host cell membrane through attachment proteins in the capsid or via glycoproteins embedded in the viral envelope . The specificity of this interaction determines the host - and the cells within the host - that can be infected by a particular virus . This can be illustrated by thinking of several keys and several locks , where each key will fit only one specific lock . Enveloped virions like HIV , the causative agent in AIDS , consist of nucleic acid (RNA in the case of HIV) and capsid proteins surrounded by a phospholipid bilayer envelope and its associated proteins . Glycoproteins embedded in the viral envelope are used to attach to host cells . Other envelope proteins are the matrix proteins that stabilize the envelope and often play a role in the assembly of progeny virions . Chicken pox , influenza , and mumps are examples of diseases caused by viruses with envelopes . Because of the fragility of the envelope , non-enveloped viruses are more resistant to changes in temperature , pH , and some disinfectants than enveloped viruses .

A virus attaches to a specific receptor site on the host cell membrane through attachment proteins in the capsid or via glycoproteins embedded in the viral envelope . The specificity of this interaction determines the host - and the cells within the host - that can be infected by a particular virus . This can be illustrated by thinking of several keys and several locks , where each key will fit only one specific lock . Viruses can be seen as obligate , intracellular parasites . A virus must attach to a living cell , be taken inside , manufacture its proteins and copy its genome , and find a way to escape the cell so that the virus can infect other cells . Viruses can infect only certain species of hosts and only certain cells within that host . Cells that a virus may use to replicate are called permissive . For most viruses , the molecular basis for this specificity is that a particular surface molecule known as the viral receptor must be found on the host cell surface for the virus to attach . Also , metabolic and host cell immune response differences seen in different cell types based on differential gene expression are a likely factor in which cells a virus may target for replication . The permissive cell must make the substances that the virus needs or the virus will not be able to replicate there .

Poglavlje 22

A large amount of available carbon is found in land plants . Plants , which are producers , use carbon dioxide from the air to synthesize carbon compounds . Related to this , one very significant source of carbon compounds is humus , which is a mixture of organic materials from dead plants and prokaryotes that have resisted decomposition . Consumers such as animals use organic compounds generated by producers and release carbon dioxide to the atmosphere . Then , bacteria and fungi , collectively called decomposers , carry out the breakdown (decomposition) of plants and animals and their organic compounds . The most important contributor of carbon dioxide to the atmosphere is microbial decomposition of dead material (dead animals , plants , and humus) that undergo respiration .

Poglavlje 25

The first fossils that show the presence of vascular tissue date to the Silurian period , about 430 million years ago . The simplest arrangement of conductive cells shows a pattern of xylem at the center surrounded by phloem . Xylem is the tissue responsible for the storage and long-distance transport of water and nutrients , as well as the transfer of water-soluble growth factors from the organs of synthesis to the target organs . The tissue consists of conducting cells , known as tracheids , and supportive filler tissue , called parenchyma . Xylem conductive cells incorporate the compound lignin into their walls , and are thus described as lignified . Lignin itself is a complex polymer that is impermeable to water and confers mechanical strength to vascular tissue . With their rigid cell walls , the xylem cells provide support to the plant and allow it to achieve impressive heights . Tall plants have a selective advantage by being able to reach unfiltered sunlight and disperse their spores or seeds further away , thus expanding their range . By growing higher than other plants , tall trees cast their shadow on shorter plants and limit competition for water and precious nutrients in the soil .

Poglavlje 29

Feathers not only act as insulation but also allow for flight , enabling the lift and thrust necessary to become airborne . The feathers on a wing are flexible , so the collective feathers move and separate as air moves through them , reducing the drag on the wing . Flight feathers are asymmetrical , which affects airflow over them and provides some of the lifting and thrusting force required for flight (Figure 29.30) . Two types of flight feathers are found on the wings , primary feathers and secondary feathers . Primary feathers are located at the tip of the wing and provide thrust . Secondary feathers are located closer to the body , attach to the forearm portion of the wing and provide lift . Contour feathers are the feathers found on the body , and they help reduce drag produced by wind resistance during flight . They create a smooth , aerodynamic surface so that air moves smoothly over the bird ' s body , allowing for efficient flight .

Poglavlje 30

Secondary tissues are either simple (composed of similar cell types) or complex (composed of different cell types) . Dermal tissue , for example , is a simple tissue that covers the outer surface of the plant and controls gas exchange . Vascular tissue is an example of a complex tissue , and is made of two specialized conducting tissues : xylem and phloem . Xylem tissue transports water and nutrients from the roots to different parts of the plant , and includes three different cell types : vessel elements and tracheids (both of which conduct water) , and xylem parenchyma . Phloem tissue , which transports organic compounds from the site of photosynthesis to other parts of the plant , consists of four different cell types : sieve cells (which conduct photosynthates) , companion cells , phloem parenchyma , and phloem fibers . Unlike xylem conducting cells , phloem conducting cells are alive at maturity . The xylem and phloem always lie adjacent to each other (Figure 30.3) . In stems , the xylem and the phloem form a structure called a vascular bundle ; in roots , this is termed the vascular stele or vascular cylinder . 30.2 Stems Learning Objectives By the end of this section , you will be able to :

Poglavlje 36

In humans , there are five primary tastes , and each taste has only one corresponding type of receptor . Thus , like olfaction , each receptor is specific to its stimulus (tastant) . Transduction of the five tastes happens through different mechanisms that reflect the molecular composition of the tastant . A salty tastant (containing NaCl) provides the sodium ions (Na +) that enter the taste neurons and excite them directly . Sour tastants are acids and belong to the thermoreceptor protein family . Binding of an acid or other sour-tasting molecule triggers a change in the ion channel and these increase hydrogen ion (H +) concentrations in the taste neurons , thus depolarizing them . Sweet , bitter , and umami tastants require a G-protein coupled receptor . These tastants bind to their respective receptors , thereby exciting the specialized neurons associated with them .

Poglavlje 37

The term " humoral " is derived from the term " humor , " which refers to bodily fluids such as blood . A humoral stimulus refers to the control of hormone release in response to changes in extracellular fluids such as blood or the ion concentration in the blood . For example , a rise in blood glucose levels triggers the pancreatic release of insulin . Insulin causes blood glucose levels to drop , which signals the pancreas to stop producing insulin in a negative feedback loop .

Poglavlje 38

Some movements that cannot be classified as gliding , angular , or rotational are called special movements . Inversion involves the soles of the feet moving inward , toward the midline of the body . Eversion is the opposite of inversion , movement of the sole of the foot outward , away from the midline of the body . Protraction is the anterior movement of a bone in the horizontal plane . Retraction occurs as a joint moves back into position after protraction . Protraction and retraction can be seen in the movement of the mandible as the jaw is thrust outwards and then back inwards . Elevation is the movement of a bone upward , such as when the shoulders are shrugged , lifting the scapulae . Depression is the opposite of elevation - movement downward of a bone , such as after the shoulders are shrugged and the scapulae return to their normal position from an elevated position . Dorsiflexion is a bending at the ankle such that the toes are lifted toward the knee . Plantar flexion is a bending at the ankle when the heel is lifted , such as when standing on the toes . Supination is the movement of the radius and ulna bones of the forearm so that the palm faces forward . Pronation is the opposite movement , in which the palm faces backward . Opposition is the movement of the thumb toward the fingers of the same hand , making it possible to grasp and hold objects . An articulation is any place at which two bones are joined . The humerus is the largest and longest bone of the upper limb and the only bone of the arm . It articulates with the scapula at the shoulder and with the forearm at the elbow . The forearm extends from the elbow to the wrist and consists of two bones : the ulna and the radius . The radius is located along the lateral (thumb) side of the forearm and articulates with the humerus at the elbow . The ulna is located on the medial aspect (pinky-finger side) of the forearm . It is longer than the radius . The ulna articulates with the humerus at the elbow . The radius and ulna also articulate with the carpal bones and with each other , which in vertebrates enables a variable degree of rotation of the carpus with respect to the long axis of the limb . The hand includes the eight bones of the carpus (wrist) , the five bones of the metacarpus (palm) , and the 14 bones of the phalanges (digits) . Each digit consists of three phalanges , except for the thumb , when present , which has only two . The upper limb contains 30 bones in three regions : the arm (shoulder to elbow) , the forearm (ulna and radius) , and the wrist and hand (Figure 38.12) .

ACh is broken down by the enzyme acetylcholinesterase (AChE) into acetyl and choline . AChE resides in the synaptic cleft , breaking down ACh so that it does not remain bound to ACh receptors , which would cause unwanted extended muscle contraction (Figure 38.38) . Acetylcholine (ACh) is a neurotransmitter released by motor neurons that binds to receptors in the motor end plate . Neurotransmitter release occurs when an action potential travels down the motor neuron ' s axon , resulting in altered permeability of the synaptic terminal membrane and an influx of calcium . The Ca^{2+} ions allow synaptic vesicles to move to and bind with the presynaptic membrane (on the neuron) , and release neurotransmitter from the vesicles into the synaptic cleft . Once released by the synaptic terminal , ACh diffuses across the synaptic cleft to the motor end plate , where it binds with ACh receptors . As a neurotransmitter binds , these ion channels open , and Na^+ ions cross the membrane into the muscle cell . This reduces the voltage difference between the inside and outside of the cell , which is called depolarization . As ACh binds at the motor end plate , this depolarization is called an end-plate potential . The depolarization then spreads along the sarcolemma , creating an action potential as sodium channels adjacent to the initial depolarization site sense the change in voltage and open . The action potential moves across the entire cell , creating a wave of depolarization . The sodium - potassium ATPase uses cellular energy to move K^+ ions inside the cell and Na^+ ions outside . This alone accumulates a small electrical charge , but a big concentration gradient . There is lots of K^+ in the cell and lots of Na^+ outside the cell . Potassium is able to leave the cell through K^+ channels that are open 90 % of the time , and it does . However , Na^+ channels are rarely open , so Na^+ remains outside the cell . When K^+ leaves the cell , obeying its concentration gradient , that effectively leaves a negative charge behind . So at rest , there is a large concentration gradient for Na^+ to enter the cell , and there is an accumulation of negative charges left behind in the cell . This is the resting membrane potential . Potential in this context means a separation of electrical charge that is capable of doing work . It is measured in volts , just like a battery . However , the transmembrane potential is considerably smaller (0.07 V); therefore , the small value is expressed as millivolts (mV) or 70 mV . Because the inside of a cell is negative compared with the outside , a minus sign signifies the excess of negative charges inside the cell , - 70 mV . If an event changes the permeability of the membrane to Na^+ ions , they will enter the cell . That will change the voltage . This is an electrical event , called an action potential , that can be used as a cellular signal . Communication occurs between nerves and muscles through neurotransmitters . Neuron action potentials cause the release of neurotransmitters from the synaptic terminal into the synaptic cleft , where they can then diffuse across the synaptic cleft and bind to a receptor molecule on the motor end plate . The motor end plate possesses junctional folds - folds in the sarcolemma that create a large surface area for the neurotransmitter to bind to receptors . The receptors are actually sodium channels that open to allow the passage of Na^+ into the cell when they receive neurotransmitter signal .

Poglavlje 42

Naïve T cells can express one of two different molecules , CD4 or CD8 , on their surface , as shown in Figure 42.11 , and are accordingly classified as $CD4^+$ or $CD8^+$ cells . These molecules are important because they regulate how a T cell will interact with and respond to an APC . Naïve $CD4^+$ cells bind APCs via their antigen-embedded MHC II molecules and are stimulated to become helper T (T_H) lymphocytes , cells that go on to stimulate B cells (or cytotoxic T cells) directly or secrete cytokines to inform more and various target cells about the pathogenic threat . In contrast , $CD8^+$ cells engage antigen-embedded MHC I molecules on APCs and are stimulated to become cytotoxic T lymphocytes (CTLs) , which directly kill infected cells by apoptosis and emit cytokines to amplify the immune response . The two populations of T cells have different mechanisms of immune protection , but both bind MHC molecules via their antigen receptors called T cell receptors (TCRs) . The CD4 or CD8 surface molecules differentiate whether the TCR will engage an MHC II or an MHC I molecule . Because they assist in binding specificity , the CD4 and CD8 molecules are described as coreceptors .

The immune system has to be regulated to prevent wasteful , unnecessary responses to harmless substances , and more importantly so that it does not attack " self . " The acquired ability to prevent an unnecessary or harmful immune response to a detected foreign substance known not to cause disease is described as immune tolerance . Immune tolerance is crucial for maintaining mucosal homeostasis given the tremendous number of foreign substances (such as food proteins) that APCs of the oral cavity , pharynx , and gastrointestinal mucosa encounter . Immune tolerance is brought about by specialized APCs in the liver , lymph nodes , small intestine , and lung that present harmless antigens to an exceptionally diverse population of regulatory T (T reg) cells , specialized lymphocytes that suppress local inflammation and inhibit the secretion of stimulatory immune factors . The combined result of T reg cells is to prevent immunologic activation and inflammation in undesired tissue compartments and to allow the immune system to focus on pathogens instead . In addition to promoting immune tolerance of harmless antigens , other subsets of T reg cells are involved in the prevention of the autoimmune response , which is an inappropriate immune response to host cells or self-antigens . Another T reg class suppresses immune responses to harmful pathogens after the infection has cleared to minimize host cell damage induced by inflammation and cell lysis .

Poglavlje 43

The mesoderm that lies on either side of the vertebrate neural tube will develop into the various connective tissues of the animal body . A spatial pattern of gene expression reorganizes the mesoderm into groups of cells called somites with spaces between them . The somites , illustrated in Figure 43.29 will further develop into the ribs , lungs , and segmental (spine) muscle . The mesoderm also forms a structure called the notochord , which is rod-shaped and forms the central axis of the animal body .

Compliance with the contraceptive method is a strong contributor to the success or failure rate of any particular method . The only method that is completely effective at preventing conception is abstinence . The choice of contraceptive method depends on the goals of the woman or couple . Tubal ligation and vasectomy are considered permanent prevention , while other methods are reversible and provide short-term contraception .

The development of multi-cellular organisms begins from a single-celled zygote , which undergoes rapid cell division to form the blastula . The rapid , multiple rounds of cell division are termed cleavage . Cleavage is illustrated in (Figure 43.24 a) . After the cleavage has produced over 100 cells , the embryo is called a blastula . The blastula is usually a spherical layer of cells (the blastoderm) surrounding a fluid-filled or yolk-filled cavity (the blastocoel) . Mammals at this stage form a structure called the blastocyst , characterized by an inner cell mass that is distinct from the surrounding blastula , shown in Figure 43.24 b . During cleavage , the cells divide without an increase in mass ; that is , one large single-celled zygote divides into multiple smaller cells . Each cell within the blastula is called a blastomere . Cleavage can take place in two ways : holoblastic (total) cleavage or meroblastic (partial) cleavage . The type of cleavage depends on the amount of yolk in the eggs . In placental mammals (including humans) where nourishment is provided by the mother ' s body , the eggs have a very small amount of yolk and undergo holoblastic cleavage . Other species , such as birds , with a lot of yolk in the egg to nourish the embryo during development , undergo meroblastic cleavage .

External fertilization in an aquatic environment protects the eggs from drying out . Broadcast spawning can result in a greater mixture of the genes within a group , leading to higher genetic diversity and a greater chance of species survival in a hostile environment . For sessile aquatic organisms like sponges , broadcast spawning is the only mechanism for fertilization and colonization of new environments . The presence of the fertilized eggs and developing young in the water provides opportunities for predation resulting in a loss of offspring . Therefore , millions of eggs must be produced by individuals , and the offspring produced through this method must mature rapidly . The survival rate of eggs produced through broadcast spawning is low . External fertilization usually occurs in aquatic environments where both eggs and sperm are released into the water . After the sperm reaches the egg , fertilization takes place . Most external fertilization happens during the process of spawning where one or several females release their eggs and the male (s) release sperm in the same area , at the same time . The release of the reproductive material may be triggered by water temperature or the length of daylight . Nearly all fish spawn , as do crustaceans (such as crabs and shrimp) , mollusks (such as oysters) , squid , and echinoderms (such as sea urchins and sea cucumbers) . Figure 43.6 shows salmon spawning in a shallow stream . Frogs , like those shown in Figure 43.7 , corals , squid , and octopuses also spawn . Pairs of fish that are not broadcast spawners may exhibit courtship behavior . This allows the female to select a particular male . The trigger for egg and sperm release (spawning) causes the egg and sperm to be placed in a small area , enhancing the possibility of fertilization .

At the onset of puberty , the hypothalamus causes the release of FSH and LH into the male system for the first time . FSH enters the testes and stimulates the Sertoli cells to begin facilitating spermatogenesis using negative feedback , as illustrated in Figure 43.14 . LH also enters the testes and stimulates the interstitial cells of Leydig to make and release testosterone into the testes and the blood . Gametogenesis , the production of sperm and eggs , takes place through the process of meiosis . During meiosis , two cell divisions separate the paired chromosomes in the nucleus and then separate the chromatids that were made during an earlier stage of the cell ' s life cycle . Meiosis produces haploid cells with half of each pair of chromosomes normally found in diploid cells . The production of sperm is called spermatogenesis and the production of eggs is called oogenesis .

Poglavlje 44

Estuaries are biomes that occur where a source of fresh water , such as a river , meets the ocean . Therefore , both fresh water and salt water are found in the same vicinity ; mixing results in a diluted (brackish) saltwater . Estuaries form protected areas where many of the young offspring of crustaceans , mollusks , and fish begin their lives . Salinity is a very important factor that influences the organisms and the adaptations of the organisms found in estuaries . The salinity of estuaries varies and is based on the rate of flow of its freshwater sources . Once or twice a day , high tides bring salt water into the estuary . Low tides occurring at the same frequency reverse the current of salt water . The ocean is the largest marine biome . It is a continuous body of salt water that is relatively uniform in chemical composition ; it is a weak solution of mineral salts and decayed biological matter . Within the ocean , coral reefs are a second kind of marine biome . Estuaries , coastal areas where salt water and fresh water mix , form a third unique marine biome .

Poglavlje 45

Iteroparity describes species that reproduce repeatedly during their lives . Some animals are able to mate only once per year , but survive multiple mating seasons . The pronghorn antelope is an example of an animal that goes into a seasonal estrus cycle (" heat "): a hormonally induced physiological condition preparing the body for successful mating (Figure 45.7 b) . Females of these species mate only during the estrus phase of the cycle . A different pattern is observed in primates , including humans and chimpanzees , which may attempt reproduction at any time during their reproductive years , even though their menstrual cycles make pregnancy likely only a few days per month during ovulation (Figure 45.7 c) .

Poglavlje 47

Contemporary societies that live close to the land often have a broad knowledge of the medicinal uses of plants growing in their area . Most plants produce secondary plant compounds , which are toxins used to protect the plant from insects and other animals that eat them , but some of which also work as medication . For centuries in Europe , older knowledge about the medical uses of plants was compiled in herbals - books that identified plants and their uses . Humans are not the only species to use plants for medicinal reasons : the great apes , orangutans , chimpanzees , bonobos , and gorillas have all been observed self-medicating with plants .

business ethics

Poglavlje 2

Rawls ' s justice theory contains three principles and five procedural steps for achieving fairness . The principles are (1) an " original position , " (2) a " veil of ignorance , " and (3) unanimity of acceptance of the original position . 61 By original position , Rawls meant something akin to Hobbes ' understanding of the state of nature , a hypothetical situation in which rational people can arrive at a contractual agreement about how resources are to be distributed in accordance with the principles of justice as fairness . This agreement was intended to reflect not present reality but a desired state of affairs among people in the community . The veil of ignorance (Figure 2.10) is a condition in which people arrive at the original position imagining they have no identity regarding age , sex , ethnicity , education , income , physical attractiveness , or other characteristics . In this way , they reduce their bias and self-interest . Last , unanimity of acceptance is the requirement that all agree to the contract before it goes into effect . Rawls hoped this justice theory would provide a minimum guarantee of rights and liberties for everyone , because no one would know , until the veil was lifted , whether they were male , female , rich , poor , tall , short , intelligent , a minority , Roman Catholic , disabled , a veteran , and so on .

Poglavlje 3

When a product does not live up to its maker ' s claims for whatever reason , the manufacturer needs to correct the problem to retain or regain customers ' trust . Without this trust , the interdependence between the company and its stakeholders can fail . By choosing to recognize and repay its customer stakeholders , Samsung acted at an ethical maximum , taking the strongest possible action to behave ethically in a given situation . An ethical minimum , or the least a company might do that complies with the law , would have been to offer the warning and nothing more . This may have been a defensible position in court , but the warning might not have reached all purchasers of the defective machine and many children could have been hurt . Samsung , based in South Korea , is a large , multinational corporation that makes a variety of products , including household appliances such as washers and dryers . When Samsung ' s washers developed a problem with the spin cycle in 2017 , the company warned customers that the machines could become unbalanced and tip over , and that children should be kept away . The problem persisted , however , and Samsung ' s responsibility and legal exposure increased . The eventual fix was to offer all owners of the particular washer model a full refund even if the customer did not have a complaint , and to offer free pick up of the machine as well . The recall covered almost three million washers , which ranged in price from $ 450 to $ 1500 . By choosing to spend billions to rectify the problem , Samsung limited its legal exposure to potential lawsuits , settlement of which would likely have far exceeded the refunds it paid . This example demonstrates the weight of the implicit social contract between a company and its stakeholders and the potential impact on the bottom line if that contract is broken .

Poglavlje 7

In general terms , the duty of loyalty means an employee is obligated to render " loyal and faithful " service to the employer , to act with " good faith , " and not to compete with but rather to advance the employer ' s interests . 1 The employee must not act in a way that benefits him - or herself (or any other third party) , especially when doing so would create a conflict of interest with the employer . 2 The common law of most states holds as a general rule that , without asking for and receiving the employer ' s consent , an employee cannot hold a second job if it would compete or conflict with the first job . Thus , although the precise boundaries of this aspect of the duty of loyalty are unclear , an employee who works in the graphic design department of a large advertising agency in all likelihood cannot moonlight on the weekend for a friend ' s small web design business . However , employers often grant permission for employees to work in positions that do not compete or interfere with their principal jobs . The graphic designer might work for a friend ' s catering business , for example , or perhaps as a wedding photographer or editor of a blog for a public interest community group .

business law i essentials

Poglavlje 1

In addition to the federal court and individual state systems , there are also a variety of mechanisms that companies can use to resolve disputes . They are collectively called alternative dispute resolution (" ADR ") , and they include mediation , settlement , and arbitration . Many states now require companies to resolve legal disputes using ADR before the initiation of any lawsuit to encourage speedy resolution , cost and time containment , and reduced judicial dockets . Traditional litigation remains an option in most cases if other efforts fail or are refused .

Poglavlje 2

Successful mediators work to immediately establish personal rapport with the disputing parties . They often have a short period of time to interact with the parties and work to position themselves as a trustworthy advisor . The Harvard Law School Program on Negotiation reports a study by mediator Peter Adler in which mediation participants remembered the mediators as " opening the room , making coffee , and getting everyone introduced . " This quote underscores the need for mediators to play a role beyond mere administrative functions . The mediator ' s conflict resolution skills are critical in guiding the parties toward reaching a resolution . Mediation is a method of dispute resolution that relies on an impartial third-party decision-maker , known as a mediator , to settle a dispute . While requirements vary by state , a mediator is someone who has been trained in conflict resolution , though often , he or she does not have any expertise in the subject matter that is being disputed . Mediation is another form of alternative dispute resolution . It is often used in attempts to resolve a dispute because it can help disagreeing parties avoid the time-consuming and expensive procedures involved in court litigation . Courts will often recommend that a plaintiff , or the party initiating a lawsuit , and a defendant , or the party that is accused of wrongdoing , attempt mediation before proceeding to trial . This recommendation is especially true for issues that are filed in small claims courts , where judges attempt to streamline dispute resolution . Not all mediators are associated with public court systems . There are many agency-connected and private mediation services that disputing parties can hire to help them potentially resolve their dispute . The American Bar Association suggests that , in addition to training courses , one of the best ways to start a private mediation business is to volunteer as a mediator . Research has shown that experience is an important factor for mediators who are seeking to cultivate sensitivity and hone their conflict resolution skills .

Poglavlje 5

Larceny and embezzlement are two forms of theft that can occur within a business . Larceny occurs when a person unlawfully takes the personal property of another person or a business . For example , if an employee takes another employee ' s computer with the intent of stealing it , he or she may be guilty of larceny . In contrast , embezzlement occurs when a person has been entrusted with an item of value and then refuses to return it or does not return the item . For example , if an employee is entrusted with the petty cash at his or her office and that person purposefully takes some of the money for himself or herself , this would be embezzlement .

Poglavlje 8

Inadequate warning against the dangers involved in using the good . Inadequate instructions on the use of the good Manufacturing defects Design defects If the buyer believes that there has been a breach of the implied warranty of merchantability , it is their responsibility to demonstrate that the good was defective , that this defect made the good not fit for purpose , and that this defect caused the plaintiff harm . Typical examples of defects are :

introduction to intellectual property

Poglavlje 1

A growing public outcry ultimately forced Elizabeth ' s successor James I to revoke these grants of trade monopolies in 1610 . In 1624 , the Statute of Monopolies formally repealed the practice and henceforth restricted patent rights patent monopolies were granted over the trade in such staples as salt , soap , starch , iron , and paper . Critics said these " enriched the monopolist and robbed the community " while doing absolutely nothing to stimulate new technology or industry .

introduction to sociology

Poglavlje 1

Another theory with a macro-level view , called conflict theory , looks at society as a competition for limited resources . Conflict theory sees society as being made up of individuals who must compete for social , political , and material resources such as political power , leisure time , money , housing , and entertainment . Social structures and organizations such as religious groups , governments , and corporations reflect this competition in their inherent inequalities . Some individuals and organizations are able to obtain and keep more resources than others . These " winners " use their power and influence to maintain their positions of power in society and to suppress the advancement of other individuals and groups . Of the early founders of sociology , Karl Marx is most closely identified with this theory . He focused on the economic conflict between different social classes . As he and Fredrick Engels famously described in their Communist Manifesto , " the history of all hitherto existing society is the history of class struggles . Freeman and slave , patrician and plebeian , lord and serf , guild-master and journeyman , in a word , oppressor and oppressed " (1848) . Marx rejected Comte's positivism . He believed that societies grew and changed as a result of the struggles of different social classes over the means of production . At the time he was developing his theories , the Industrial Revolution and the rise of capitalism led to great disparities in wealth between the owners of the factories and workers . Capitalism , an economic system characterized by private or corporate ownership of goods and the means to produce them , grew in many nations .

Poglavlje 4

Émile Durkheim and Functionalism As a functionalist , Émile Durkheim ' s (1858 - 1917) perspective on society stressed the necessary interconnectivity of all of its elements . To Durkheim , society was greater than the sum of its parts . He asserted that individual behavior was not the same as collective behavior , and that studying collective behavior was quite different from studying an individual ' s actions . Durkheim called the communal beliefs , morals , and attitudes of a society the collective conscience . In his quest to understand what causes individuals to act in similar and predictable ways , he wrote , " If I do not submit to the conventions of society , if in my dress I do not conform to the customs observed in my country and in my class , the ridicule I provoke , the social isolation in which I am kept , produce , although in an attenuated form , the same effects as punishment " (Durkheim 1895) . Durkheim also believed that social integration , or the strength of ties that people have to their social groups , was a key factor in social life .

Poglavlje 5

For example , in the United States , schools have built a sense of competition into the way grades are awarded and the way teachers evaluate students (Bowles and Gintis 1976) . When children participate in a relay race or a math contest , they learn that there are winners and losers in society . When children are required to work together on a project , they practice teamwork with other people in cooperative situations . The hidden curriculum prepares children for the adult world . Children learn how to deal with bureaucracy , rules , expectations , waiting their turn , and sitting still for hours during the day . Schools in different cultures socialize children differently in order to prepare them to function well in those cultures . The latent functions of teamwork and dealing with bureaucracy are features of American culture . Most American children spend about seven hours a day , 180 days a year , in school , which makes it hard to deny the importance school has on their socialization (U . S . Department of Education 2004) . Students are not only in school to study math , reading , science , and other subjects - the manifest function of this system . Schools also serve a latent function in society by socializing children into behaviors like teamwork , following a schedule , and using textbooks .

Poglavlje 9

Social stratification systems determine social position based on factors like income , education , and occupation . Sociologists use the term status consistency to describe the consistency , or lack thereof , of an individual ' s rank across these factors . Caste systems correlate with high status consistency , whereas the more flexible class system has lower status consistency . Sociologists use the term social stratification to describe the system of social standing . Social stratification refers to a society ' s categorization of its people into rankings of socioeconomic tiers based on factors like wealth , income , race , education , and power .

Poglavlje 11

For several decades , Mexican workers crossed the long border into America , both legally and illegally , to work in the fields that provided produce for the developing United States . Western growers needed a steady supply of labor , and the 1940s and 1950s saw the official federal Bracero Program (bracero is Spanish for strong-arm) that offered protection to Mexican guest workers . Interestingly , 1954 also saw the enactment of " Operation Wetback , " which deported thousands of illegal Mexican workers . From these examples , we can see that the U . S . treatment of immigration from Mexico has been ambivalent at best . Not only are there wide differences among the different origins that make up the Hispanic American population , there are also different names for the group itself . The 2010 U . S . Census states that " Hispanic " or " Latino " refers to a person of Cuban , Mexican , Puerto Rican , South or Central American , or other Spanish culture or origin regardless of race . " There have been some disagreements over whether Hispanic or Latino is the correct term for a group this diverse , and whether it would be better for people to refer to themselves as being of their origin specifically , for example , Mexican American or Dominican American . This section will compare the experiences of Mexican Americans and Cuban Americans .

Poglavlje 13

Certainly , as boomers age , they will put increasing burdens on the entire U . S . health care system . A study from 2008 indicates that medical schools are not producing enough medical professionals who specialize in treating geriatric patients (Gerontological Society of America 2008) . However , other studies indicate that aging boomers will bring economic growth to the health care industries , particularly in areas like pharmaceutical manufacturing and home health care services (Bierman 2011) . Further , some argue that many of our medical advances of the past few decades are a result of boomers ' health requirements . Unlike the elderly of previous generations , boomers do not expect that turning 65 means their active lives are over . They are not willing to abandon work or leisure activities , but they may need more medical support to keep living vigorous lives . This desire of a large group of over - 65 - year-olds wanting to continue with a high activity level is driving innovation in the medical industry (Shaw) .

Two factors contribute significantly to this country ' s aging prison population . One is the tough-on-crime reforms of the 1980s and 1990s , when mandatory minimum sentencing and " three strikes " policies sent many people to jail for 30 years to life , even when the third strike was a relatively minor offense (Leadership Conference N . d .) . Many of today ' s elderly prisoners are those who were incarcerated 30 years ago for life sentences . The other factor influencing today ' s aging prison population is the aging of the overall population . As discussed in the section on aging in the United States , the percentage of people over 65 is increasing each year due to rising life expectancies and the aging of the baby boom generation .

Poglavlje 14

Family is , indeed , a subjective concept , but it is a fairly objective fact that family (whatever one ' s concept of it may be) is very important to Americans . In a 2010 survey by Pew Research Center in Washington , D . C . , 76 percent of adults surveyed stated that family is " the most important " element of their life - just one percent said it was " not important " (Pew Research Center 2010) . It is also very important to society . President Ronald Regan notably stated , " The family has always been the cornerstone of American society . Our families nurture , preserve , and pass on to each succeeding generation the values we share and cherish , values that are the foundation of our freedoms " (Lee 2009) . While the design of the family may have changed in recent years , the fundamentals of emotional closeness and support are still present . Most responders to the Pew survey stated that their family today is at least as close (45 percent) or closer (40 percent) than the family with which they grew up (Pew Research Center 2010) .

Poglavlje 16

Education also fulfills latent functions . As you well know , much goes on in a school that has little to do with formal education . For example , you might notice an attractive fellow student when he gives a particularly interesting answer in class - catching up with him and making a date speaks to the latent function of courtship fulfilled by exposure to a peer group in the educational setting . Education also provides one of the major methods used by people for upward social mobility . This function is referred to as social placement . College and graduate schools are viewed as vehicles for moving students closer to the careers that will give them the financial freedom and security they seek . As a result , college students are often more motivated to study areas that they believe will be advantageous on the social ladder . A student might value business courses over a class in Victorian poetry because she sees business class as a stronger vehicle for financial success . There are several major manifest functions associated with education . The first is socialization . Beginning in preschool and kindergarten , students are taught to practice various societal roles . The French sociologist Émile Durkheim (1858 - 1917) , who established the academic discipline of sociology , characterized schools as " socialization agencies that teach children how to get along with others and prepare them for adult economic roles " (Durkheim 1898) . Indeed , it seems that schools have taken on this responsibility in full .

Until the 1954 Brown v . Board of Education ruling , schools had operated under the precedent set by Plessy v . Ferguson in 1896 , which allowed racial segregation in schools and private businesses (the case dealt specifically with railroads) and introduced the much maligned phrase " separate but equal " into the United States lexicon . The 1954 Brown v . Board decision overruled this , declaring that state laws that had established separate schools for black and white students were , in fact , unequal and unconstitutional .

Poglavlje 17

Modern-day life is not without a multitude of political conflicts : discontents in Egypt overthrow dictator Hosni Mubarak , disenchanted American Tea Partiers call for government realignment , and Occupy Wall Street protesters decry corporate greed . Indeed , the study of any given conflict offers a window of insight into the social structure of its surrounding culture , as well as insight into the larger human condition

Poglavlje 18

Governments tried to protect their share of the markets by developing a system called mercantilism . Mercantilism is an economic policy based on accumulating silver and gold by controlling colonial and foreign markets through taxes and other charges . The resulting restrictive practices and exacting demands included monopolies , bans on certain goods , high tariffs , and exclusivity requirements . Mercantilistic governments also promoted manufacturing and , with the ability to fund technological improvements , they helped create the equipment that led to the Industrial Revolution .

Poglavlje 19

Among whites , Hispanic whites received 60 percent inferior care of measures compared to non-Hispanic whites (Agency for Health Research and Quality 2010) . When considering access to care , the figures were comparable . Health by Race and Ethnicity When looking at the social epidemiology of the United States , it is hard to miss the disparities among races . The discrepancy between black and white Americans shows the gap clearly ; IN 2008 , the average life expectancy for white males was approximately five years longer than for black males : 75.9 compared to 70.9 . An even stronger disparity was found in 2007 : the infant mortality rate for blacks was nearly twice that of whites at 13.2 compared to 5.6 per 1,000 live births (U . S . Census Bureau 2011) . According to a report from the Henry J . Kaiser Foundation (2007) , African Americans also have higher incidence of several other diseases and causes of mortality , from cancer to heart disease to diabetes . In a similar vein , it is important to note that ethnic minorities , including Mexican Americans and Native Americans , also have higher rates of these diseases and causes of mortality than whites . Globally , the rate of mortality for children under five was 60 per 1,000 live births . In low-income countries , however , that rate is almost double at 117 per 1,000 live births . In high-income countries , that rate is significantly lower than seven per 1,000 live births .

Poglavlje 20

Social Policy and Debate Suburbs Are Not All White Picket Fences : The Banlieues of Paris What makes a suburb a suburb ? Simply , a suburb is a community surrounding a city . But when you picture a suburb in your mind , your image may vary widely depending on which nation you call home . In the United States , most consider the suburbs home to upper and middle class people with private homes . In other countries , like France , the suburbs - - or " banlieues " - - are synonymous with housing projects and impoverished communities . In fact , the banlieues of Paris are notorious for their ethnic violence and crime , with higher unemployment and more residents living in poverty than in the city center . Further , the banlieues have a much higher immigrant population , which in Paris is mostly Arabic and African immigrants . This contradicts the clichéd American image of a typical white-picket-fence suburb .

Poglavlje 21

McCarthy and Zald (1977) conceptualize resource mobilization theory as a way to explain movement success in terms of its ability to acquire resources and mobilize individuals . For example , PETA , a social movement organization , is in competition with Greenpeace and the Animal Liberation Front (ALF) , two other social movement organizations . Taken together , along with all other social movement organizations working on animals rights issues , these similar organizations constitute a social movement industry . Multiple social movement industries in a society , though they may have widely different constituencies and goals , constitute a society's social movement sector . Every social movement organization (a single social movement group) within the social movement sector is competing for your attention , your time , and your resources . The chart below shows the relationship between these components . The conflict perspective focuses on the creation and reproduction of inequality . Someone applying the conflict perspective would likely be interested in how social movements are generated through systematic inequality , and how social change is constant , speedy , and unavoidable . In fact , the conflict that this perspective sees as inherent in social relations drives social change . For example , the National Association for the Advancement of Colored People (NAACP) was founded in 1908 . Partly created in response to the horrific lynchings occurring in the southern United States , the organization fought to secure the constitutional rights guaranteed in the 13th , 14th , and 15th amendments , which established an end to slavery , equal protection under the law , and universal male suffrage (NAACP 2011) . While those goals have been achieved , the organization remains active today , continuing to fight against inequalities in civil rights and to remedy discriminatory practices . What do Arab Spring , Occupy Wall Street , People for the Ethical Treatment of Animals (PETA) , the anti-globalization movement , and the Tea Party have in common ? Not much , you might think . But although they may be left-wing or right-wing , radical or conservative , highly organized or very diffused , they are all examples of social movements .

microbiology

Poglavlje 4

The green nonsulfur bacteria are similar to green sulfur bacteria but they use substrates other than sulfides for oxidation . Chloroflexus is an example of a green nonsulfur bacterium . It often has an orange color when it grows in the dark , but it becomes green when it grows in sunlight . It stores bacteriochlorophyll in chlorosomes , similar to Chlorobium , and performs anoxygenic photosynthesis , using organic sulfites (low concentrations) or molecular hydrogen as electron donors , so it can survive in the dark if oxygen is available . Chloroflexus does not have flagella but can glide , like Cytophaga . It grows at a wide range of temperatures , from 35 ° C to 70 ° C , thus can be thermophilic .

Poglavlje 5

The dinoflagellates and stramenopiles fall within the Chromalveolata . The dinoflagellates are mostly marine organisms and are an important component of plankton . They have a variety of nutritional types and may be phototrophic , heterotrophic , or mixotrophic . Those that are photosynthetic use chlorophyll a , chlorophyll c_2 , and other photosynthetic pigments (Figure 5.35) . They generally have two flagella , causing them to whirl (in fact , the name dinoflagellate comes from the Greek word for " whirl " : dini) . Some have cellulose plates forming a hard outer covering , or theca , as armor . Additionally , some dinoflagellates produce neurotoxins that can cause paralysis in humans or fish . Exposure can occur through contact with water containing the dinoflagellate toxins or by feeding on organisms that have eaten dinoflagellates .

Poglavlje 7

Isomers that differ in the spatial arrangements of atoms are called stereoisomers ; one unique type is enantiomers . The properties of enantiomers were originally discovered by Louis Pasteur in 1848 while using a microscope to analyze crystallized fermentation products of wine . Enantiomers are molecules that have the characteristic of chirality , in which their structures are nonsuperimposable mirror images of each other . Chirality is an important characteristic in many biologically important molecules , as illustrated by the examples of structural differences in the enantiomeric forms of the monosaccharide glucose or the amino acid alanine (Figure 7.5) .

The next level of protein organization is the tertiary structure , which is the large-scale three-dimensional shape of a single polypeptide chain . Tertiary structure is determined by interactions between amino acid residues that are far apart in the chain . A variety of interactions give rise to protein tertiary structure , such as disulfide bridge s , which are bonds between the sulfhydryl (- SH) functional groups on amino acid side groups ; hydrogen bonds ; ionic bonds ; and hydrophobic interactions between nonpolar side chains . All these interactions , weak and strong , combine to determine the final three-dimensional shape of the protein and its function (Figure 7.21) .

Poglavlje 9

$N_n = N_0 2^n$ $N_n = N_0 2^n$ The number of cells increases exponentially and can be expressed as 2^n , where n is the number of generations . If cells divide every 30 minutes , after 24 hours , 48 divisions would have taken place . If we apply the formula 2^n , where n is equal to 48 , the single cell would give rise to 2^{48} or 281,474 , 976,710 , 656 cells at 48 generations (24 hours) . When dealing with such huge numbers , it is more practical to use scientific notation . Therefore , we express the number of cells as 2.8×10^{14} cells .

Measuring dry weight of a culture sample is another indirect method of evaluating culture density without directly measuring cell counts . The cell suspension used for weighing must be concentrated by filtration or centrifugation , washed , and then dried before the measurements are taken . The degree of drying must be standardized to account for residual water content . This method is especially useful for filamentous microorganisms , which are difficult to enumerate by direct or viable plate count .

Poglavlje 10

During DNA packaging , DNA-binding proteins called histones perform various levels of DNA wrapping and attachment to scaffolding proteins . The combination of DNA with these attached proteins is referred to as chromatin . In eukaryotes , the packaging of DNA by histones may be influenced by environmental factors that affect the presence of methyl groups on certain cytosine nucleotides of DNA . The influence of environmental factors on DNA packaging is called epigenetics . Epigenetics is another mechanism for regulating gene expression without altering the sequence of nucleotides . Epigenetic changes can be maintained through multiple rounds of cell division and , therefore , can be heritable .

Poglavlje 12

Another method of transfecting plants involves shuttle vectors , plasmids that can move between bacterial and eukaryotic cells . The tumor-inducing (T i) plasmids originating from the bacterium Agrobacterium tumefaciens are commonly used as shuttle vectors for incorporating genes into plants (Figure 12.12) . In nature , the T i plasmids of A . tumefaciens cause plants to develop tumors when they are transferred from bacterial cells to plant cells . Researchers have been able to manipulate these naturally occurring plasmids to remove their tumor-causing genes and insert desirable DNA fragments . The resulting recombinant T i plasmids can be transferred into the plant genome through the natural transfer of T i plasmids from the bacterium to the plant host . Once inside the plant host cell , the gene of interest recombines into the plant cell ' s genome .

Molecules with complementary sticky ends can easily anneal , or form hydrogen bonds between complementary bases , at their sticky ends . The annealing step allows hybridization of the single-stranded overhangs . Hybridization refers to the joining together of two complementary single strands of DNA . Blunt ends can also attach together , but less efficiently than sticky ends due to the lack of complementary overhangs facilitating the process . In either case , ligation by DNA ligase can then rejoin the two sugar-phosphate backbones of the DNA through covalent bonding , making the molecule a continuous double strand . In 1972 , Paul Berg , a Stanford biochemist , was the first to produce a recombinant DNA molecule using this technique , combining the SV40 monkey virus with E . coli bacteriophage lambda to create a hybrid .

Poglavlje 15

To cause disease , a pathogen must successfully achieve four steps or stages of pathogenesis : exposure (contact) , adhesion (colonization) , invasion , and infection . The pathogen must be able to gain entry to the host , travel to the location where it can establish an infection , evade or overcome the host ' s immune response , and cause damage (i . e . , disease) to the host . In many cases , the cycle is completed when the pathogen exits the host and is transmitted to a new host .

Poglavlje 18

Helper T cells can only be activated by APCs presenting processed foreign epitopes in association with MHC II . The first step in the activation process is TCR recognition of the specific foreign epitope presented within the MHC II antigen-binding cleft . The second step involves the interaction of CD4 on the helper T cell with a region of the MHC II molecule separate from the antigen-binding cleft . This second interaction anchors the MHC II-TCR complex and ensures that the helper T cell is recognizing both the foreign (" nonself ") epitope and " self " antigen of the APC ; both recognitions are required for activation of the cell . In the third step , the APC and T cell secrete cytokines that activate the helper T cell . The activated helper T cell then proliferates , dividing by mitosis to produce clonal naïve helper T cells that differentiate into subtypes with different functions (Figure 18.17) .

Poglavlje 19

Treatments of type III hypersensitivities include preventing further exposure to the antigen and the use of anti-inflammatory drugs . Some conditions can be resolved when exposure to the antigen is prevented . Anti-inflammatory corticosteroid inhalers can also be used to diminish inflammation to allow lung lesions to heal . Systemic corticosteroid treatment , oral or intravenous , is also common for type III hypersensitivities affecting body systems . Treatment of hypersensitivity pneumonitis includes avoiding the allergen , along with the possible addition of prescription steroids such as prednisone to reduce inflammation .

Poglavlje 23

Microscopic evaluation of wet mounts is an inexpensive and convenient method of diagnosis , but the sensitivity of this method is low (Figure 23.24) . Nucleic acid amplification testing (NAAT) is preferred due to its high sensitivity . Using wet mounts and then NAAT for those who initially test negative is one option to improve sensitivity . Samples may be obtained for NAAT using urine , vaginal , or endocervical specimens for women and with urine and urethral swabs for men . It is also possible to use other methods such as the OSOM Trichomonas Rapid Test (an immunochromatographic test that detects antigen) and a DNA probe test for multiple species associated with vaginitis (the Affirm VPII Microbial Identification Test discussed in section 23.5) . 16 T . vaginalis is sometimes detected on a Pap test , but this is not considered diagnostic due to high rates of false positives and negatives . The recommended treatment for trichomoniasis is oral metronidazole or tinidazole . Sexual partners should be treated as well . 16 Association of Public Health Laboratories . " Advances in Laboratory Detection of Trichomonas vaginalis , " 2013 . http://www.aphl.org/AboutAPHL/publications/Documents/ID_2013August_Advances-in-Laboratory-Detection-of-Trichomonas-vaginalis.pdf .

Clinical Focus Resolution Vaginal candidiasis is generally treated using topical antifungal medications such as butoconazole , miconazole , clotrimazole , ticonozole , nystatin , or oral fluconazole . However , it is important to be careful in selecting a treatment for use during pregnancy . Nadia ' s doctor recommended treatment with topical clotrimazole . This drug is classified as a category B drug by the FDA for use in pregnancy , and there appears to be no evidence of harm , at least in the second or third trimesters of pregnancy . Based on Nadia ' s particular situation , her doctor thought that it was suitable for very short-term use even though she was still in the first trimester . After a seven-day course of treatment , Nadia ' s yeast infection cleared . She continued with a normal pregnancy and delivered a healthy baby eight months later . Topical antifungal medications for vaginal candidiasis include butoconazole , miconazole , clotrimazole , tioconazole , and nystatin . Oral treatment with fluconazole can be used . There are often no clear precipitating factors for infection , so prevention is difficult .

Poglavlje 24

The constant movement of materials through the gastrointestinal tract also helps to move transient pathogens out of the body . In fact , feces are composed of approximately 25 % microbes , 25 % sloughed epithelial cells , 25 % mucus , and 25 % digested or undigested food . Finally , the normal microbiota provides an additional barrier to infection via a variety of mechanisms . For example , these organisms outcompete potential pathogens for space and nutrients within the intestine . This is known as competitive exclusion . Members of the microbiota may also secrete protein toxins known as bacteriocins that are able to bind to specific receptors on the surface of susceptible bacteria . The environment of most of the GI tract is harsh , which serves two purposes : digestion and immunity . The stomach is an extremely acidic environment (pH 1.5 - 3.5) due to the gastric juices that break down food and kill many ingested microbes ; this helps prevent infection from pathogens . The environment in the small intestine is less harsh and is able to support microbial communities . Microorganisms present in the small intestine can include lactobacilli , diptherioids and the fungus Candida . On the other hand , the large intestine (colon) contains a diverse and abundant microbiota that is important for normal function . These microbes include Bacteriodetes (especially the genera Bacteroides and Prevotella) and Firmicutes (especially members of the genus Clostridium) . Methanogenic archaea and some fungi are also present , among many other species of bacteria . These microbes all aid in digestion and contribute to the production of feces , the waste excreted from the digestive tract , and flatus , the gas produced from microbial fermentation of undigested food . They can also produce valuable nutrients . For example , lactic acid bacteria such as bifidobacteria can synthesize vitamins , such as vitamin B12 , folate , and riboflavin , that humans cannot synthesize themselves . E . coli found in the intestine can also break down food and help the body produce vitamin K , which is important for blood coagulation .

The signs and symptoms of norovirus infection are similar to those for rotavirus , with watery diarrhea , mild cramps , and fever . Additionally , these viruses sometimes cause projectile vomiting . The illness is usually relatively mild , develops 12 to 48 hours after exposure , and clears within a couple of days without treatment . However , dehydration may occur .

Poglavlje 25

The circulatory (or cardiovascular) system is a closed network of organs and vessels that moves blood around the body (Figure 25.2) . The primary purposes of the circulatory system are to deliver nutrients , immune factors , and oxygen to tissues and to carry away waste products for elimination . The heart is a four-chambered pump that propels the blood throughout the body . Deoxygenated blood enters the right atrium through the superior vena cava and the inferior vena cava after returning from the body . The blood next passes through the tricuspid valve to enter the right ventricle . When the heart contracts , the blood from the right ventricle is pumped through the pulmonary arteries to the lungs . There , the blood is oxygenated at the alveoli and returns to the heart through the pulmonary veins . The oxygenated blood is received at the left atrium and proceeds through the mitral valve to the left ventricle . When the heart contracts , the oxygenated blood is pumped throughout the body via a series of thick-walled vessels called arteries . The first and largest artery is called the aorta . The arteries sequentially branch and decrease in size (and are called arterioles) until they end in a network of smaller vessels called capillaries . The capillary beds are located in the interstitial spaces within tissues and release nutrients , immune factors , and oxygen to those tissues . The capillaries connect to a series of vessels called venules , which increase in size to form the vein s . The veins join together into larger vessels as they transfer blood back to the heart . The largest veins , the superior and inferior vena cava , return the blood to the right atrium .

Poglavlje 26

Cryptococcus neoformans is a fungal pathogen that can cause meningitis . This yeast is commonly found in soils and is particularly associated with pigeon droppings . It has a thick capsule that serves as an important virulence factor , inhibiting clearance by phagocytosis . Most C . neoformans cases result in subclinical respiratory infections that , in healthy individuals , generally resolve spontaneously with no long-term consequences (see Respiratory Mycoses) . In immunocompromised patients or those with other underlying illnesses , the infection can progress to cause meningitis and granuloma formation in brain tissues . Cryptococcus antigens can also serve to inhibit cell-mediated immunity and delayed-type hypersensitivity .

principles of accounting, volume 1: financial accounting

Poglavlje 1

External users are those outside of the organization who use the financial information to make decisions or to evaluate an entity ' s performance . For example , investors , financial analysts , loan officers , governmental auditors , such as IRS agents , and an assortment of other stakeholders are classified as external users , while still having an interest in an organization ' s financial information . (Stakeholders are addressed in greater detail in Explain Why Accounting Is Important to Business Stakeholders .)

Poglavlje 2

In this example using a fictitious company , Cheesy Chuck ' s , we began with the account balances and demonstrated how to prepare the financial statements for the month of June , the first month of operations for the business . It will be helpful to revisit the process by summarizing the information we started with and how that information was used to create the four financial statements : income statement , statement of owner ' s equity , balance sheet , and statement of cash flows .

Poglavlje 3

Advertising is an expense of doing business . You have incurred more expenses , so you want to increase an expense account . Expense accounts increase with debit entries . Debit advertising expense . Net income reported on the income statement flows into the statement of retained earnings . If a business has net income (earnings) for the period , then this will increase its retained earnings for the period . This means that revenues exceeded expenses for the period , thus increasing retained earnings . If a business has net loss for the period , this decreases retained earnings for the period . This means that the expenses exceeded the revenues for the period , thus decreasing retained earnings .

Poglavlje 5

Why was income summary not used in the dividends closing entry ? Dividends are not an income statement account . Only income statement accounts help us summarize income , so only income statement accounts should go into income summary .

The last step for the month of August is step 9 , preparing the post-closing trial balance . The post-closing trial balance should only contain permanent account information . No temporary accounts should appear on this trial balance . Clip ' em Cliff ' s post-closing trial balance is presented in Figure 5.27 . The process of preparing the post-closing trial balance is the same as you have done when preparing the unadjusted trial balance and adjusted trial balance . Only permanent account balances should appear on the post-closing trial balance . These balances in post-closing T-accounts are transferred over to either the debit or credit column on the post-closing trial balance . When all accounts have been recorded , total each column and verify the columns equal each other .

Poglavlje 6

Operating expenses are daily operational costs not associated with the direct selling of products or services . Operating expenses are broken down into selling expenses (such as advertising and marketing expenses) and general and administrative expenses (such as office supplies expense , and depreciation of office equipment) . Deducting the operating expenses from gross margin produces income from operations . A multi-step income statement is more detailed than a simple income statement . Because of the additional detail , it is the option selected by many companies whose operations are more complex . Each revenue and expense account is listed individually under the appropriate category on the statement . The multi-step statement separates cost of goods sold from operating expenses and deducts cost of goods sold from net sales to obtain a gross margin .

Poglavlje 7

We use accounting information to make decisions about the business . Computer applications now provide so much data that data analytics is one of the newest career areas in business . Universities are beginning to offer degrees in data analysis . Software companies have created different applications to analyze data including SAS , Apache Hadoop , Apache Spark , SPSS , RapidMiner , Power BI , ACL , IDEA , and many more to help companies discover useful information from the transactions that occur . Big data refers to the availability of large amounts of data from various sources , including the internet . For example , social media sites contain tremendous amounts of data that marketing companies analyze to determine how popular a product is , and how best to market it . There is so much data to analyze that new ways of mining it for predictive value have evolved .

Poglavlje 11

The expense recognition principle that requires that the cost of the asset be allocated over the asset ' s useful life is the process of depreciation . For example , if we buy a delivery truck to use for the next five years , we would allocate the cost and record depreciation expense across the entire five-year period . The calculation of the depreciation expense for a period is not based on anticipated changes in the fair market value of the asset ; instead , the depreciation is based on the allocation of the cost of owning the asset over the period of its useful life . Depreciation is the process of allocating the cost of a tangible asset over its useful life , or the period of time that the business believes it will use the asset to help generate revenue . This process will be described in Explain and Apply Depreciation Methods to Allocate Capitalized Costs .

Poglavlje 14

Stock can be issued in exchange for cash , property , or services provided to the corporation . For example , an investor could give a delivery truck in exchange for a company ' s stock . Another investor could provide legal fees in exchange for stock . The general rule is to recognize the assets received in exchange for stock at the asset ' s fair market value .

Regulate securities markets Enforce federal securities laws Facilitate capital information Inform and protect investors The Securities and Exchange Commission (SEC) (www.sec.gov) is a government agency that regulates large and small public corporations . Its mission is " to protect investors , maintain fair , orderly , and efficient markets , and facilitate capital formation . " 3 The SEC identifies these as its five primary responsibilities : 3 U . S . Securities and Exchange Commission . " What We Do . " June 10 , 2013 . https://www.sec.gov/Article/whatwedo.html

principles of accounting, volume 2: managerial accounting

Poglavlje 1

The control function helps to determine the courses of action that are taken in the implementation of a plan by helping to define and administer the steps of the plan . Essentially , the control function facilitates coordination of the plan within the organization . It is through the system of controls that the actual results of decisions made in implementing a plan can be identified and measured . Managerial accounting not only helps to determine and design control measures , it also assists by providing performance reports and control reports that focus on variances between the planned objective performance and the actual performance . Control is achieved through effective feedback , or information that is used to assess a process . Feedback allows management to evaluate the results , determine whether progress is being made , or determine whether corrective measures need to be taken . This evaluation is in the next management function .

Poglavlje 2

A merchandising firm is one of the most common types of businesses . A merchandising firm is a business that purchases finished products and resells them to consumers . Consider your local grocery store or retail clothing store . Both of these are merchandising firms . Often , merchandising firms are referred to as resellers or retailers since they are in the business of reselling a product to the consumer at a profit .

Poglavlje 4

Indirect labor records are also maintained through time tickets , although such work is not directly traceable to a specific job . The difference between direct labor and indirect labor is that the indirect labor records the debit to manufacturing overhead while the credit is to factory wages payable . Job order cost systems maintain the actual direct materials and direct labor for each individual job . Since production consists of overhead - indirect materials , indirect labor , and other overhead - we need a methodology for applying that overhead . Unfortunately , the nature of indirect material , indirect labor , and other overhead expenses makes it impossible to determine the exact amount of overhead for each specific job . For example , how do you know the cost of electricity and heat for manufacturing one job ? And , if you did , is it fair to say products manufactured in January are more expensive than the same product manufactured in March because of heat expense ?

Poglavlje 8

Once a company determines a standard cost , they can then evaluate any variances . A variance is the difference between a standard cost and actual performance . There are favorable and unfavorable variances . A favorable variance involves spending less , or using less , than the anticipated or estimated standard . An unfavorable variance involves spending more , or using more , than the anticipated or estimated standard . Before determining whether the variance is favorable or unfavorable , it is often helpful for the company to determine why the variance exists .

Poglavlje 9

New businesses , for example , are often centralized . When a business first opens , it is common for the owner (s) to be highly involved in the day-to-day operations . In addition , the small size of a new business allows the owner to have a high level of involvement in both the daily and the strategic decisions of the business . Daily decisions are ongoing , immediate decisions that must be made in order to effectively and efficiently meet the needs of the organization ' s customers . Strategic decisions , on the other hand , are made fairly infrequently and involve long-term goals of the organization . Being actively involved in the business allows new business owners to gain experience in all aspects of the business so that they can get a sense of the patterns of the daily operations and the decisions that need to be made . For example , the owner can be involved in determining the number of workers needed to meet the day ' s production goal . Having too many workers would be inefficient and require the company to incur unnecessary expenses . Having too few workers , on the other hand , may result in inferior quality of products , missed shipments , or lost sales .

Poglavlje 10

The Robinson-Patman Act prevents large retailers from purchasing goods in bulk at a greater discount than smaller retailers are able to obtain them . It helps keep competition fair between large and small businesses and is sometimes called the " Anti-Chain Store Act . " Read the LegalDictionary.net full definition and example of the Robinson-Patman Act to learn more .

Poglavlje 12

In addition to being timely , performance measures need to be applied or measured consistently . The accounting variables or other measures that are used to evaluate a manager should be measured the same way from period to period . For example , if a performance measure includes some form of income , such as operating income , then that measure should be used each time and not replaced with another income measure for the current measurement cycle (usually one year) . If , upon further analysis , it seems that net income is a better measure to use in the evaluation of a manager , then the new measure can be implemented during the next measurement cycle . When measures are changed , it is imperative that the manager being evaluated is aware of the measurement change , as this may affect his or her decision-making . The idea is to keep the targets stable for a period . Otherwise , the measurements might be inconsistent , and thus misleading . A good performance measurement plan would include the manager ' s input in the design discussion . Not only does this help to ensure that the plan is clear to all parties involved in the process , it also helps to motivate managers . Rather than being told what goals are to be met , managers will be more motivated to achieve the goals if they have input into the process , the goals to be reached , and the measurements or metrics being used .

These areas were chosen by Kaplan and Norton because the success of a company is dependent on how it performs financially , which is directly related to the company ' s internal operations , how the customer perceives and interacts with the company , and the direction in which the company is headed . The use of the balanced scorecard allows the company to take a stakeholder perspective as compared to a stockholder perspective . Stockholders are the owners of the company stock and often are most concerned with the profitability of the company and thus focus primarily on financial results . Stakeholders are people who are affected by the decisions made by a company , such as investors , creditors , managers , regulators , employees , customers , suppliers , and even laypeople who are concerned about whether or not the company is a good world citizen . This is why social responsibility factors are sometimes included in balanced scorecards . To understand where these types of factors might fit in a balanced scorecard framework , let ' s look at the four sections or categories of a balanced scorecard .

Therefore , a balanced scorecard evaluates employees on an assortment of quantitative factors , or metrics based on financial information , and qualitative factors , or those based on nonfinancial information , in several significant areas . The quantitative or financial measurements tend to emphasize past results , often based on their financial statements , while the qualitative or nonfinancial measurements center on current results or activities , with the intent to evaluate activities that will influence future financial performance .

psychology

Poglavlje 3

The sympathetic nervous system is activated when we are faced with stressful or high-arousal situations . The activity of this system was adaptive for our ancestors , increasing their chances of survival . Imagine , for example , that one of our early ancestors , out hunting small game , suddenly disturbs a large bear with her cubs . At that moment , his body undergoes a series of changes - a direct function of sympathetic activation - preparing him to face the threat . His pupils dilate , his heart rate and blood pressure increase , his bladder relaxes , his liver releases glucose , and adrenaline surges into his bloodstream . This constellation of physiological changes , known as the fight or flight response , allows the body access to energy reserves and heightened sensory capacity so that it might fight off a threat or run away to safety .

Poglavlje 4

Drug withdrawal includes a variety of negative symptoms experienced when drug use is discontinued . These symptoms usually are opposite of the effects of the drug . For example , withdrawal from sedative drugs often produces unpleasant arousal and agitation . In addition to withdrawal , many individuals who are diagnosed with substance use disorders will also develop tolerance to these substances . Psychological dependence , or drug craving , is a recent addition to the diagnostic criteria for substance use disorder in DSM - 5 . This is an important factor because we can develop tolerance and experience withdrawal from any number of drugs that we do not abuse . In other words , physical dependence in and of itself is of limited utility in determining whether or not someone has a substance use disorder . The fifth edition of the Diagnostic and Statistical Manual of Mental Disorders , Fifth Edition (DSM - 5) is used by clinicians to diagnose individuals suffering from various psychological disorders . Drug use disorders are addictive disorders , and the criteria for specific substance (drug) use disorders are described in DSM - 5 . A person who has a substance use disorder often uses more of the substance than they originally intended to and continues to use that substance despite experiencing significant adverse consequences . In individuals diagnosed with a substance use disorder , there is a compulsive pattern of drug use that is often associated with both physical and psychological dependence . Physical dependence involves changes in normal bodily functions - the user will experience withdrawal from the drug upon cessation of use . In contrast , a person who has psychological dependence has an emotional , rather than physical , need for the drug and may use the drug to relieve psychological distress . Tolerance is linked to physiological dependence , and it occurs when a person requires more and more drug to achieve effects previously experienced at lower doses . Tolerance can cause the user to increase the amount of drug used to a dangerous level - even to the point of overdose and death .

Poglavlje 6

Like Tolman , whose experiments with rats suggested a cognitive component to learning , psychologist Albert Bandura ' s ideas about learning were different from those of strict behaviorists . Bandura and other researchers proposed a brand of behaviorism called social learning theory , which took cognitive processes into account . According to Bandura , pure behaviorism could not explain why learning can take place in the absence of external reinforcement . He felt that internal mental states must also have a role in learning and that observational learning involves much more than imitation . In imitation , a person simply copies what the model does . Observational learning is much more complex . According to Lefrançois (2012) there are several ways that observational learning can occur :

Both reflexes and instincts help an organism adapt to its environment and do not have to be learned . For example , every healthy human baby has a sucking reflex , present at birth . Babies are born knowing how to suck on a nipple , whether artificial (from a bottle) or human . Nobody teaches the baby to suck , just as no one teaches a sea turtle hatchling to move toward the ocean .

Poglavlje 9

In the discussion of biopsychology earlier in the book , you learned about genetics and DNA . A mother and father ' s DNA is passed on to the child at the moment of conception . Conception occurs when sperm fertilizes an egg and forms a zygote (Figure 9.7) . A zygote begins as a one-cell structure that is created when a sperm and egg merge . The genetic makeup and sex of the baby are set at this point . During the first week after conception , the zygote divides and multiplies , going from a one-cell structure to two cells , then four cells , then eight cells , and so on . This process of cell division is called mitosis . Mitosis is a fragile process , and fewer than one-half of all zygotes survive beyond the first two weeks (Hall , 2004) . After 5 days of mitosis there are 100 cells , and after 9 months there are billions of cells . As the cells divide , they become more specialized , forming different organs and body parts . In the germinal stage , the mass of cells has yet to attach itself to the lining of the mother ' s uterus . Once it does , the next stage begins .

Poglavlje 11

One of Mischel ' s most notable contributions to personality psychology was his ideas on self-regulation . According to Lecci & Magnavita (2013) , " Self-regulation is the process of identifying a goal or set of goals and , in pursuing these goals , using both internal (e . g . , thoughts and affect) and external (e . g . , responses of anything or anyone in the environment) feedback to maximize goal attainment " (p . 6.3) . Self-regulation is also known as will power . When we talk about will power , we tend to think of it as the ability to delay gratification . For example , Bettina ' s teenage daughter made strawberry cupcakes , and they looked delicious . However , Bettina forfeited the pleasure of eating one , because she is training for a 5K race and wants to be fit and do well in the race . Would you be able to resist getting a small reward now in order to get a larger reward later ? This is the question Mischel investigated in his now-classic marshmallow test .

Poglavlje 12

As discussed previously , social roles are defined by a culture ' s shared knowledge of what is expected behavior of an individual in a specific role . This shared knowledge comes from social norms . A social norm is a group ' s expectation of what is appropriate and acceptable behavior for its members - how they are supposed to behave and think (Deutsch & Gerard , 1955 ; Berkowitz , 2004) . How are we expected to act ? What are we expected to talk about ? What are we expected to wear ? In our discussion of social roles we noted that colleges have social norms for students ' behavior in the role of student and workplaces have social norms for employees ' behaviors in the role of employee . Social norms are everywhere including in families , gangs , and on social media outlets . What are some social norms on Facebook ?

Poglavlje 13

One form of sexual harassment is called quid pro quo . Quid pro quo means you give something to get something , and it refers to a situation in which organizational rewards are offered in exchange for sexual favors . Quid pro quo harassment is often between an employee and a person with greater power in the organization . For example , a supervisor might request an action , such as a kiss or a touch , in exchange for a promotion , a positive performance review , or a pay raise . Another form of sexual harassment is the threat of withholding a reward if a sexual request is refused . Hostile environment sexual harassment is another type of workplace harassment . In this situation , an employee experiences conditions in the workplace that are considered hostile or intimidating . For example , a work environment that allows offensive language or jokes or displays sexually explicit images . Isolated occurrences of these events do not constitute harassment , but a pattern of repeated occurrences does . In addition to violating organizational policies against sexual harassment , these forms of harassment are illegal . Title VII of the Civil Rights Act of 1964 makes it illegal to treat individuals unfavorably because of their race or color of their skin : An employer cannot discriminate based on skin color , hair texture , or other immutable characteristics , which are traits of an individual that are fundamental to her identity , in hiring , benefits , promotions , or termination of employees . The Pregnancy Discrimination Act of 1978 amends the Civil Rights Act ; it prohibits job (e . g . , employment , pay , and termination) discrimination of a woman because she is pregnant as long as she can perform the work required .

A significant portion of I-O research focuses on management and human relations . Douglas McGregor (1960) combined scientific management (a theory of management that analyzes and synthesizes workflows with the main objective of improving economic efficiency , especially labor productivity) and human relations into the notion of leadership behavior . His theory lays out two different styles called Theory X and Theory Y . In the Theory X approach to management , managers assume that most people dislike work and are not innately self-directed . Theory X managers perceive employees as people who prefer to be led and told which tasks to perform and when . Their employees have to be watched carefully to be sure that they work hard enough to fulfill the organization ' s goals . Theory X workplaces will often have employees punch a clock when arriving and leaving the workplace : Tardiness is punished . Supervisors , not employees , determine whether an employee needs to stay late , and even this decision would require someone higher up in the command chain to approve the extra hours . Theory X supervisors will ignore employees ' suggestions for improved efficiency and reprimand employees for speaking out of order . These supervisors blame efficiency failures on individual employees rather than the systems or policies in place . Managerial goals are achieved through a system of punishments and threats rather than enticements and rewards . Managers are suspicious of employees ' motivations and always suspect selfish motivations for their behavior at work (e . g . , being paid is their sole motivation for working) .

Poglavlje 15

The disturbances lead to significant distress or disability in one ' s life . A person ' s inner experiences and behaviors are considered to reflect a psychological disorder if they cause the person considerable distress , or greatly impair his ability to function as a normal individual (often referred to as functional impairment , or occupational and social impairment) . As an illustration , a person ' s fear of social situations might be so distressing that it causes the person to avoid all social situations (e . g . , preventing that person from being able to attend class or apply for a job) .

A child with ADHD shows a constant pattern of inattention and / or hyperactive and impulsive behavior that interferes with normal functioning (APA , 2013) . Some of the signs of inattention include great difficulty with and avoidance of tasks that require sustained attention (such as conversations or reading) , failure to follow instructions (often resulting in failure to complete school work and other duties) , disorganization (difficulty keeping things in order , poor time management , sloppy and messy work) , lack of attention to detail , becoming easily distracted , and forgetfulness . Hyperactivity is characterized by excessive movement , and includes fidgeting or squirming , leaving one ' s seat in situations when remaining seated is expected , having trouble sitting still (e . g . , in a restaurant) , running about and climbing on things , blurting out responses before another person ' s question or statement has been completed , difficulty waiting one ' s turn for something , and interrupting and intruding on others . Frequently , the hyperactive child comes across as noisy and boisterous . The child ' s behavior is hasty , impulsive , and seems to occur without much forethought ; these characteristics may explain why adolescents and young adults diagnosed with ADHD receive more traffic tickets and have more automobile accidents than do others (Thompson , Molina , Pelham , & Gnagy , 2007) . Diego likely has attention deficit / hyperactivity disorder (ADHD) . The symptoms of this disorder were first described by Hans Hoffman in the 1920s . While taking care of his son while his wife was in the hospital giving birth to a second child , Hoffman noticed that the boy had trouble concentrating on his homework , had a short attention span , and had to repeatedly go over easy homework to learn the material (Jellinek & Herzog , 1999) . Later , it was discovered that many hyperactive children - those who are fidgety , restless , socially disruptive , and have trouble with impulse control - also display short attention spans , problems with concentration , and distractibility . By the 1970s , it had become clear that many children who display attention problems often also exhibit signs of hyperactivity . In recognition of such findings , the DSM-III (published in 1980) included a new disorder : attention deficit disorder with and without hyperactivity , now known as attention deficit / hyperactivity disorder (ADHD) .

Poglavlje 16

Starting in 1954 and gaining popularity in the 1960s , antipsychotic medications were introduced . These proved a tremendous help in controlling the symptoms of certain psychological disorders , such as psychosis . Psychosis was a common diagnosis of individuals in mental hospitals , and it was often evidenced by symptoms like hallucinations and delusions , indicating a loss of contact with reality . Then in 1963 , Congress passed and John F . Kennedy signed the Mental Retardation Facilities and Community Mental Health Centers Construction Act , which provided federal support and funding for community mental health centers (National Institutes of Health , 2013) . This legislation changed how mental health services were delivered in the United States . It started the process of deinstitutionalization , the closing of large asylums , by providing for people to stay in their communities and be treated locally . In 1955 , there were 558,239 severely mentally ill patients institutionalized at public hospitals (Torrey , 1997) . By 1994 , by percentage of the population , there were 92 % fewer hospitalized individuals (Torrey , 1997) .

u.s. history

Poglavlje 3

At the same time , European goods had begun to change Native life radically . In the 1500s , some of the earliest objects Europeans introduced to Native Americans were glass beads , copper kettles , and metal utensils . Native people often adapted these items for their own use . For example , some cut up copper kettles and refashioned the metal for other uses , including jewelry that conferred status on the wearer , who was seen as connected to the new European source of raw materials .

Puritan New England differed in many ways from both England and the rest of Europe . Protestants emphasized literacy so that everyone could read the Bible . This attitude was in stark contrast to that of Catholics , who refused to tolerate private ownership of Bibles in the vernacular . The Puritans , for their part , placed a special emphasis on reading scripture , and their commitment to literacy led to the establishment of the first printing press in English America in 1636 . Four years later , in 1640 , they published the first book in North America , the Bay Psalm Book . As Calvinists , Puritans adhered to the doctrine of predestination , whereby a few " elect " would be saved and all others damned . No one could be sure whether they were predestined for salvation , but through introspection , guided by scripture , Puritans hoped to find a glimmer of redemptive grace . Church membership was restricted to those Puritans who were willing to provide a conversion narrative telling how they came to understand their spiritual estate by hearing sermons and studying the Bible .

Poglavlje 4

Though Cromwell enjoyed widespread popularity at first , over time he appeared to many in England to be taking on the powers of a military dictator . Dissatisfaction with Cromwell grew . When he died in 1658 and control passed to his son Richard , who lacked the political skills of his father , a majority of the English people feared an alternate hereditary monarchy in the making . They had had enough and asked Charles II to be king . In 1660 , they welcomed the son of the executed king Charles I back to the throne to resume the English monarchy and bring the interregnum to an end (Figure 4.3) . The return of Charles II is known as the Restoration . When Charles II ascended the throne in 1660 , English subjects on both sides of the Atlantic celebrated the restoration of the English monarchy after a decade of living without a king as a result of the English Civil Wars . Charles II lost little time in strengthening England ' s global power . From the 1660s to the 1680s , Charles II added more possessions to England ' s North American holdings by establishing the Restoration colonies of New York and New Jersey (taking these areas from the Dutch) as well as Pennsylvania and the Carolinas . In order to reap the greatest economic benefit from England ' s overseas possessions , Charles II enacted the mercantilist Navigation Acts , although many colonial merchants ignored them because enforcement remained lax .

That year , thirteen fires broke out in the city , one of which reduced the colony ' s Fort George to ashes . Ever fearful of an uprising among enslaved New Yorkers , the city ' s White residents spread rumors that the fires were part of a massive revolt in which enslaved people would murder White people , burn the city , and take over the colony . The Stono Rebellion was only a few years in the past , and throughout British America , fears of similar incidents were still fresh . Searching for solutions , and convinced their captives were the principal danger , nervous British authorities interrogated almost two hundred enslaved people and accused them of conspiracy . Rumors that Roman Catholics had joined the suspected conspiracy and planned to murder Protestant inhabitants of the city only added to the general hysteria . Very quickly , two hundred people were arrested , including a large number of the city ' s enslaved population .

Poglavlje 6

As the British had hoped , large numbers of Loyalists helped ensure the success of the southern strategy , and thousands of enslaved individuals seeking freedom arrived to aid Cornwallis ' s army . However , the war turned in the Americans ' favor in 1781 . General Greene realized that to defeat Cornwallis , he did not have to win a single battle . So long as he remained in the field , he could continue to destroy isolated British forces . Greene therefore made a strategic decision to divide his own troops to wage war - and the strategy worked . American forces under General Daniel Morgan decisively beat the British at the Battle of Cowpens in South Carolina . General Cornwallis now abandoned his strategy of defeating the backcountry rebels in South Carolina . Determined to destroy Greene ' s army , he gave chase as Greene strategically retreated north into North Carolina . At the Battle of Guilford Courthouse in March 1781 , the British prevailed on the battlefield but suffered extensive losses , an outcome that paralleled the Battle of Bunker Hill nearly six years earlier in June 1775 . The disaster at Charleston led the Continental Congress to change leadership by placing General Horatio Gates in charge of American forces in the South . However , General Gates fared no better than General Lincoln ; at the Battle of Camden , South Carolina , in August 1780 , Cornwallis forced General Gates to retreat into North Carolina . Camden was one of the worst disasters suffered by American armies during the entire Revolutionary War . Congress again changed military leadership , this time by placing General Nathanael Greene (Figure 6.14) in command in December 1780 . By 1778 , the war had turned into a stalemate . Although some in Britain , including Prime Minister Lord North , wanted peace , King George III demanded that the colonies be brought to obedience . To break the deadlock , the British revised their strategy and turned their attention to the southern colonies , where they could expect more support from Loyalists . The southern colonies soon became the center of the fighting . The southern strategy brought the British success at first , but thanks to the leadership of George Washington and General Nathanael Greene and the crucial assistance of French forces , the Continental Army defeated the British at Yorktown , effectively ending further large-scale operations during the war .

Paine ' s pamphlet rejected the monarchy , calling King George III a " royal brute " and questioning the right of an island (England) to rule over America . In this way , Paine helped to channel colonial discontent toward the king himself and not , as had been the case , toward the British Parliament - a bold move that signaled the desire to create a new political order disavowing monarchy entirely . He argued for the creation of an American republic , a state without a king , and extolled the blessings of republicanism , a political philosophy that held that elected representatives , not a hereditary monarch , should govern states . The vision of an American republic put forward by Paine included the idea of popular sovereignty : citizens in the republic would determine who would represent them , and decide other issues , on the basis of majority rule . Republicanism also served as a social philosophy guiding the conduct of the Patriots in their struggle against the British Empire . It demanded adherence to a code of virtue , placing the public good and community above narrow self-interest .

Poglavlje 14

By offering two candidates for president , the Democrats gave the Republicans an enormous advantage . Also hoping to prevent a Republican victory , pro-Unionists from the border states organized the Constitutional Union Party and put up a fourth candidate , John Bell , for president , who pledged to end slavery agitation and preserve the Union but never fully explained how he ' d accomplish this objective . In a pro-Lincoln political cartoon of the time (Figure 14.20) , the presidential election is presented as a baseball game . Lincoln stands on home plate . A skunk raises its tail at the other candidates . Holding his nose , southern Democrat John Breckinridge holds a bat labeled " Slavery Extension " and declares " I guess I ' d better leave for Kentucky , for I smell something strong around here , and begin to think , that we are completely skunk ' d . " The Democratic nominating convention met in April 1860 in Charleston , South Carolina . However , it broke up after northern Democrats , who made up a majority of delegates , rejected Jefferson Davis ' s efforts to protect slavery in the territories . These northern Democratic delegates knew that supporting Davis on this issue would be very unpopular among the people in their states . A second conference , held in Baltimore , further illustrated the divide within the Democratic Party . Northern Democrats nominated Stephen Douglas , while southern Democrats , who met separately , put forward Vice President John Breckinridge from Kentucky . The Democratic Party had fractured into two competing sectional factions . The election of 1860 threatened American democracy when the elevation of Abraham Lincoln to the presidency inspired secessionists in the South to withdraw their states from the Union . Lincoln ' s election owed much to the disarray in the Democratic Party . The Dred Scott decision and the Freeport Doctrine had opened up huge sectional divisions among Democrats . Though Brown did not intend it , his raid had furthered the split between northern and southern Democrats . Fire-Eaters vowed to prevent a northern Democrat , especially Illinois ' s Stephen Douglas , from becoming their presidential candidate . These proslavery zealots insisted on a southern Democrat .

As president , Taylor sought to defuse the sectional controversy as much as possible , and , above all else , to preserve the Union . Although Taylor was born in Virginia before relocating to Kentucky and enslaved more than one hundred people by the late 1840s , he did not push for slavery ' s expansion into the Mexican Cession . However , the California Gold Rush made California ' s statehood into an issue demanding immediate attention . In 1849 , after California residents adopted a state constitution prohibiting slavery , President Taylor called on Congress to admit California and New Mexico as free states , a move that infuriated southern defenders of slavery who argued for the right to bring their enslaved property wherever they chose . Taylor , who did not believe slavery could flourish in the arid lands of the Mexican Cession because the climate prohibited plantation-style farming , proposed that the Wilmot Proviso be applied to the entire area .

Poglavlje 15

The other major Union campaigns of 1864 were more successful and gave President Lincoln the advantage that he needed to win reelection in November . In August 1864 , the Union navy captured Mobile Bay . General Sherman invaded the Deep South , advancing slowly from Tennessee into Georgia , confronted at every turn by the Confederates , who were commanded by Johnston . When President Davis replaced Johnston with General John B . Hood , the Confederates made a daring but ultimately costly direct attack on the Union army that failed to drive out the invaders . Atlanta fell to Union forces on September 2 , 1864 . The fall of Atlanta held tremendous significance for the war-weary Union and helped to reverse the North ' s sinking morale . In keeping with the logic of total war , Sherman ' s forces cut a swath of destruction to Savannah . On Sherman ' s March to the Sea , the Union army , seeking to demoralize the South , destroyed everything in its path , despite strict instructions regarding the preservation of civilian property . Although towns were left standing , houses and barns were burned . Homes were looted , food was stolen , crops were destroyed , orchards were burned , and livestock was killed or confiscated . Savannah fell on December 21 , 1864 - a Christmas gift for Lincoln , Sherman proclaimed . In 1865 , Sherman ' s forces invaded South Carolina , capturing Charleston and Columbia . In Columbia , the state capital , the Union army burned slaveholders ' homes and destroyed much of the city . From South Carolina , Sherman ' s force moved north in an effort to join Grant and destroy Lee ' s army .

Poglavlje 21

Progressive groups made tremendous strides on issues involving democracy , efficiency , and social justice . But they found that their grassroots approach was ill-equipped to push back against the most powerful beneficiaries of growing inequality , economic concentration , and corruption - big business . In their fight against the trusts , Progressives needed the leadership of the federal government , and they found it in Theodore Roosevelt in 1901 , through an accident of history . Muckrakers drew public attention to some of the most glaring inequities and scandals that grew out of the social ills of the Gilded Age and the hands-off approach of the federal government since the end of Reconstruction . These writers by and large addressed a White , middle-class and elite , native-born audience , even though Progressive movements and organizations involved a diverse range of Americans . What united these Progressives beyond their different backgrounds and causes was a set of uniting principles , however . Most strove for a perfection of democracy , which required the expansion of suffrage to worthy citizens and the restriction of political participation for those considered " unfit " on account of health , education , or race . Progressives also agreed that democracy had to be balanced with an emphasis on efficiency , a reliance on science and technology , and deference to the expertise of professionals . They repudiated party politics but looked to government to regulate the modern market economy . And they saw themselves as the agents of social justice and reform , as well as the stewards and guides of workers and the urban poor . Often , reformers ' convictions and faith in their own expertise led them to dismiss the voices of the very people they sought to help .

Poglavlje 22

Taft ' s policies , although not as based on military aggression as his predecessors , did create difficulties for the United States , both at the time and in the future . Central America ' s indebtedness would create economic concerns for decades to come , as well as foster nationalist movements in countries resentful of American ' s interference . In Asia , Taft ' s efforts to mediate between China and Japan served only to heighten tensions between Japan and the United States . Furthermore , it did not succeed in creating a balance of power , as Japan ' s reaction was to further consolidate its power and reach throughout the region . Of key interest to Taft was the debt that several Central American nations still owed to various countries in Europe . Fearing that the debt holders might use the monies owed as leverage to use military intervention in the Western Hemisphere , Taft moved quickly to pay off these debts with U . S . dollars . Of course , this move made the Central American countries indebted to the United States , a situation that not all nations wanted . When a Central American nation resisted this arrangement , however , Taft responded with military force to achieve the objective . This occurred in Nicaragua when the country refused to accept American loans to pay off its debt to Great Britain . Taft sent a warship with marines to the region to pressure the government to agree . Similarly , when Mexico considered the idea of allowing a Japanese corporation to gain significant land and economic advantages in its country , Taft urged Congress to pass the Lodge Corollary , an addendum to the Roosevelt Corollary , stating that no foreign corporation - other than American ones - could obtain strategic lands in the Western Hemisphere .

Poglavlje 25

Although it was a relatively quiet period for U . S . diplomacy , Hoover did help to usher in a period of positive relations , specifically with several Latin American neighbors . This would establish the basis for Franklin Roosevelt ' s " Good Neighbor " policy . After a goodwill tour of Central American countries immediately following his election in 1928 , Hoover shaped the subsequent Clark Memorandum - released in 1930 - which largely repudiated the previous Roosevelt Corollary , establishing a basis for unlimited American military intervention throughout Latin America . To the contrary , through the memorandum , Hoover asserted that greater emphasis should be placed upon the older Monroe Doctrine , in which the U . S . pledged assistance to her Latin American neighbors should any European powers interfere in Western Hemisphere affairs . Hoover further strengthened relations to the south by withdrawing American troops from Haiti and Nicaragua . Additionally , he outlined with Secretary of State Henry Stimson the Hoover-Stimson Doctrine , which announced that the United States would never recognize claims to territories seized by force (a direct response to the recent Japanese invasion of Manchuria) . Other diplomatic overtures met with less success for Hoover . Most notably , in an effort to support the American economy during the early stages of the Depression , the president signed into law the Smoot-Hawley Tariff in 1930 . The law , which raised tariffs on thousands of imports , was intended to increase sales of American-made goods , but predictably angered foreign trade partners who in turn raised their tariffs on American imports , thus shrinking international trade and closing additional markets to desperate American manufacturers . As a result , the global depression worsened further . A similar attempt to spur the world economy , known as the Hoover Moratorium , likewise met with great opposition and little economic benefit . Issued in 1931 , the moratorium called for a halt to World War I reparations to be paid by Germany to France , as well as forgiveness of Allied war debts to the U . S .